
Pattern Arithmetic: Typed Algebra of Bounded Units on a Phase-Colored Sphere

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Abstract

Counting answers *how many?*, but it is only one arithmetic. Colors and musical notes are also units of truth: combining them yields a hue or a chord, not a larger integer. This paper builds one such arithmetic. Its unit is the *information atom*, a typed point $x = (r, \hat{n}, \chi, \tau)$ on a unit sphere $\mathcal{S} \subset \mathbb{R}^3$ ¹: strength $r \in [0, 1]$, direction $\hat{n} \in S^2$, internal phase-color $\chi \in [0, 2\pi)$, and sort τ . An atom may also hold several such vectors, in groups called cores. Six verbs act on atoms. Bundle ($\uplus$) lists who is present and does not double identical states. Collapse ($\oplus$) joins atoms keeping every contribution, and UnCollapse ($\ominus$) takes one core back. Mix ($\boxplus$) reads a bowl as a phasor sum and has no inverse, since a Mixed result no longer determines its sources. Transform (T) walks an order, so the same ingredients may form different expressions. Offset reuses a walk. Assembly ($\boxtimes$) relates two units without fusing them, through an external seat, and units left alone descend its energy until a stated threshold freezes them in one camp or several. Words and numbers share the sphere under different encodings and meet by order and typed action, not by blending. The claim is narrow: a combination rule should match the unit. It is not a theory of language, physics, biology, or computation.

1 Introduction

Summary. Counting answers the question *how many?* Other kinds of information combine under other rules: red and green light make yellow, two notes make a chord, a tone and its opposite phase make silence. This paper takes that observation as a thesis. Wherever a domain supplies a unit of truth and a small set of rules by which such units combine, the unit and its rules together form an arithmetic of that domain.

The paper builds one such arithmetic. Its unit is the *information atom*, also called the unit: a strength, a direction on a sphere, a phase, and a sort that says which operations

¹Three dimensions is the standing assumption throughout — every axiom, proof, and worked example below uses $\mathbb{R}^3$ unless a section states otherwise. Section 4.4 extends T to $S^{N_{\text{dim}}-1}$ for $N_{\text{dim}} \neq 3$; that is a named exception, not a change to the default, as is the general-dimension form of offset’s transport in §5.1.

may act on it. An atom may hold several such vectors, in groups called cores. Six verbs act on atoms, set side by side in the table below. Each is stated as an axiom or a rule, with worked numbers. Meta vectors carry time, location, link and the like, outside the atom's content.

The claim is narrow. A rule for combining should match the unit. The paper offers no theory of language, physics or biology, and it names its refusals as plainly as its results. The order of the paper is axioms, then verbs, then sorts, then examples. The sections that follow give the background and motivation in full.

At a glance. The unit is one typed point $x = (r, \hat{n}, \chi, \tau)$: strength $r \in [0, 1]$, direction $\hat{n} \in S^2$, phase $\chi \in [0, 2\pi)$, sort τ . The six verbs, each answering one question:

Verb	Question	What it does	Takes back	Where
Bundle $\uplus$	What is present?	Union of a bowl; sort-blind; drops order and count; $A \uplus A = A$	Remove $\setminus$	Axiom 4 (§2.5)
Collapse $\oplus$	What is combined, with nothing discarded?	Keeps every core; any sorts; $A \oplus A \neq A$	UnCollapse $\ominus$, one core	§3.3
Mix $\boxplus$	What composite results?	Phasor sum at one shared direction; lossy; refused across differing directions	none: UnMix is impossible	Axiom 5 (§2.6)
Transform T	What happens, in what order?	Turns $\hat{n}$ by πr_w about the actor's axis and adds phase; $T_a T_b \neq T_b T_a$	inverse actor w^*	§4.2
Offset	What walk is reused?	Copies a relation between two seats to a third; does not name the result	none stated	§5
Assembly $\bowtie$	How do two units relate while both still sit?	Relates intact units by an interaction energy; never repels	none	Axiom 6 (§2.9)

A bowl of units is read by a path to a result, and a result may be seated again. A unit does not know what it constitutes: a seated word does not know the poem, and the path knows it. Each verb name is reserved, the way a keyword is reserved in a programming language: once given a meaning here, it is not reused elsewhere in this paper for an unrelated or informal sense.

Background and motivation. Ordinary arithmetic uses the integer 1 as a unit of truth. The identity $1 = 1$ says that two tokens are of the same kind. The rule $1 + 1 = 2$ says that combining two such tokens produces a larger quantitative state. That algebra answers the question *how many?*

It is not the only algebra. A color and a musical note are also units of truth, but their combination does not enlarge a pile. Under additive light, red and green yield yellow. Two copies of the same yellow, once brightness is normalized, remain yellow. Idempotence is a property of $\uplus$ alone: $A \uplus A = A$ always, whereas $A \boxplus A$ grows strength to $r' = \min(1, 2r)$ at unchanged hue, and returns the same yellow only after that normalization (Axiom 5). Two notes yield a chord, not a larger integer; a tone plus its opposite phase yields silence. In each case the word “add” names a different rule, and therefore a different kind of unit.

Thesis (an arithmetic of units of truth). *Mathematics need not be performed on numbers alone. Wherever a domain supplies a unit of truth —*

the smallest thing it treats as one — and a small set of rules by which such units combine, obeyed consistently within that domain, the unit and its rules together form an arithmetic of that domain. Counting is one instance: the unit is a token and the rule is +. Such rules are seldom invented. They are discovered, as the outcomes a domain returns every time the same conditions are repeated and it is left to settle toward stability.

That is what an arithmetic *is*. What it is *for* is a result reached by a method that is consistent, reliable, and repeatable under the same conditions. Counting is one such method; it is not the only one. A hue, a chord, or a well-formed word can be such a result on another unit. This paper supplies the unit and the verbs. It does not, by itself, issue a verdict: a bowl may be mixed, walked, or assembled and brought to rest. Saying what that state means is a later reading, not a seventh verb here.

The technique is not new. Wherever a domain already treats something as one unit and combines those units under a small repeatable rule, people have been doing an arithmetic of that domain. Pāṇini’s grammar is the sharp linguistic case: roots and operations generate Sanskrit word-forms; commentators have called that engine mathematical, but only loosely. The best-known such claim is Ingerman’s 1967 proposal to rename Backus–Naur notation *Pāṇini–Backus form*, on the grounds that Pāṇini’s own rule notation is structurally equivalent in power; later readers have cautioned that the comparison simplifies features — rule-ordering, blocking, scope — a bare rewrite-grammar reading misses. The kinship worth keeping does not need that debate settled. A pratyāhāra, one of the Śiva Sūtras’ own shorthand sound-classes (for instance $IK = \{i, u, \text{ṛ}, \text{ḷ}\}$), lets one rule govern a whole class of sounds at once — the same move this paper makes with a sort τ . And where two of Pāṇini’s own rules apply to one input, his meta-rule for resolving it, *vipratishedha* (“in a conflict, the later rule prevails, unless a more specific rule stands as a stated exception to a general one”), is a precedence discipline, not a shared mechanism. Unit plus rule, same conditions, same well-formed whole — not an identity with $\uplus$ or T . Placing a chair at a table is the weakest picture and is kept only as a reminder that the same two units, combined the same way, can settle to the same usable whole.

Examples are everywhere once looked for. Light combines additively and pigment subtractively: one domain, color, carries two arithmetics, because it is the rule set and not the domain that defines one. Notes combine into chords; a plate driven at the same frequency settles into the same nodal figure (§2.7); water freezing under the same conditions returns the same six-fold symmetry. The oldest instances are older than any notation for them. Atoms combine only in the proportions their valences allow, with mass and charge conserved across the reaction. Conserved quantities in particle physics, electric charge among them, add across every interaction, and some combinations are forbidden outright. In biology, an engulfed bacterium and its host became one cell, and the mitochondrion still carries a genome of its own: two units joined into a stable whole without either being dissolved — the shape of assembly ([Axiom 6](#)), not of a blend.

These domains are cited as evidence that such arithmetics exist, not as domains this paper models; [Axiom 0](#) decides which sorts earn a geometric reading here, and most physical domains do not. Ice is the plainest case: its six-fold growth is a genuine arithmetic of its own domain, recorded by Axiom 0 as the snowflake’s native law, and it is exactly the kind refused a map onto this sphere. This paper states its own rules as axioms rather than discovering them. The second half of the thesis appears in it only in miniature: in the enclosure of [Section 9](#), units left under one fixed rule settle toward equilibrium and are read at a stated checkpoint.

Summing these separate realities into one shared register is a harder problem than cataloguing them, and this paper does not claim to have solved it. The difficulty shows even inside the one sort admitted here: Mix itself refuses two states that do not already share a direction ([Mix across differing directions: refused](#)), settled by declining rather than by finding a rule, and combining across sorts entirely — a color with a conserved charge, a chord with a valence bond — has no stated rule here at all ([Axiom 0](#)). Arriving at higher-order realities this way is worth attempting regardless. An informational atom, as defined next, is not offered as the one unit these domains secretly share; it is one further candidate in the same lineage as a pratyāhāra, a valence bond, a conserved charge — tested here on a narrow slice of the larger question, not a claim to have already answered it.

Physics shows the same difficulty at the largest scale. Quantum mechanics and general relativity each describe their own domain with great precision, and each speaks in its own terms, so joining them into one account remains an open question. Pattern Arithmetic does not address that question, and nothing here applies to either theory. The refusal to Mix across differing directions (§2.7) is a small case of the same problem, settled by declining rather than by a found rule, and it is a reason to begin at a narrow scale and test it honestly.

When the object of interest is a structured whole rather than a count, a hue, or a pitch, the natural unit is a *pattern*. This paper represents one pattern by one typed point on a unit sphere $\mathcal{S} \subset \mathbb{R}^3$, not by a field painted over the globe:

- $r \in [0, 1]$ is the strength of the unit, capped so that a state remains a single unit of truth — an activation, not a global probability density;
- $\hat{n} \in S^2$, or equivalently (θ, ϕ) , is the direction of the pattern on the sphere;
- $\chi \in [0, 2\pi)$ is an internal phase-color, so two identities may occupy the same direction without overwriting one another;
- τ is the sort (`word`, `num`, `color`, ...): a legality tag, not a fifth geometric angle.

Pattern Arithmetic is the algebra of these units. Bundle does not double identical states ($A \uplus A = A$). A Bundled unit may be removed ($\{A\} \setminus A = \emptyset$). Rotations of the sphere preserve the unit. Opposite-phase Mix is a later reading; it is not $\uplus$.

A single numerical picture is enough to see the difference from counting. Take two unit states at the same place, one with phase 0° (red) and one with phase 120° (green). Counting would return 2. Phase-aware combination returns a single unit whose blended phase is 60° (yellow). That is this paper's color inclusion — phasor Mix on a toy chart, not CIE or linear RGB ([Axiom 0](#), [Axiom 5](#)). The same two strengths with opposite phase (0° and 180°) return the null state, black. Two units of truth may become a new identity, a stronger mark of the same identity, or none.

The working order of the paper is not that Mix alone. One pipeline is

$$\text{Bundle (bowl)} \longrightarrow T \text{ (path)} \longrightarrow \text{Result}.$$

The other is assemble ($\bowtie$) then rest of seats. Both use the same unit. Mixing is a later reading of a bowl, not a third spine. Collecting red and green is not yet yellow, and not yet a poem. Yellow is what you get if the later reading is Mix ($\boxplus$). Red and green may equally well walk a T_{poem} path; the result need not be yellow — though a given

poem may still arrive there. The later act, not the shells, chooses the result. A verse is another path. Each word of “Twinkle, twinkle, little star...” is a unit state on the same sphere. Bundle yields the bag of distinct words (**twinkle** appears twice in the line and only once in the bowl). The poem is the path that walks those states in order, including the repeated **twinkle**. The bag is not the song. Numericity is not stored by $\uplus$. Two identical units Bundle to one address; that is the point of $A \uplus A = A$. How many times the address is *visited* lives in the path (the tape says **twinkle** twice). How many as a *count* lives in **num** mode, where $1 + 1 + 1 + 1 = 4$ and four tokens of 1 are not Bundled. Strength r is a poor substitute for either: a second **twinkle** cannot honestly raise r past 1, and “four ones” is schoolbook addition, not a fuller unit of truth. The same bowl may be read more than one way. Loading is not meaning.

The sphere is the stage. It is not the deepest claim. Strip away geometry, phase, rotations, and the language examples, and one proposition remains: different kinds of information require different notions of combination, and an operation should be read by what informational property it *preserves* — and what it is allowed to throw away.

$$1 + 1 = 2, \quad \text{red} \boxplus \text{green} = \text{yellow}, \quad A \uplus A = A, \quad T_a(T_b(x)) \neq T_b(T_a(x)).$$

These are not alternative tricks for the same plus sign. They answer different questions, as the table of the six verbs above sets out.

Division as deconvolution or signal unmixing is not among these verbs; shown ill-posed for this algebra, not merely unbuilt (Section 7.2). Inverse rotation of T is well-defined. A Mix inverse — UnMix, $\boxminus$ — is not (Section 3.2).

A pattern can be treated as one unit. Identity and multiplicity come apart: $\uplus$ keeps kind and discards count; the path keeps visits; **num** keeps quantity. Composition and transformation are different verbs. Sorts may share a stage without sharing an operator. The geometry that follows is only a language in which those verbs can be written. It does not mint the relations; a lexicon or a learning rule must still seat Sun and Day. Section 1.4 keeps that claim narrow.

1.1 The Limitations of Discretized Matrix Grids

A discrete matrix grid is the wrong box for such a unit. Traditional matrix representations of continuous state spaces suffer from two severe structural bottlenecks:

- **Memory Scaling Bottlenecks:** A voxel cube of resolution N_{vox} stores $O(N_{\text{vox}}^3)$ cells, growing steeply with resolution.
- **Signal Identity Destruction:** Mapping multiple overlapping variables to identical spatial coordinates often results in the destructive overwriting of underlying signal identities.

1.2 The Phase-Colored Unit Sphere Framework

To resolve these grid constraints, the sphere is offered as one bounded object that can carry overlapping identities by direction and phase, without enlarging a lattice. Its four coordinates, strength r , direction $\hat{n}$, phase χ and sort τ , are those given above.

1.3 Principal Contributions & High-Level Intuition

The principal contributions of Pattern Arithmetic are:

- **A typed unit:** one atom $x = (r, \hat{n}, \chi, \tau)$ on a bounded sphere, not a voxel cell and not a global probability field.
- **Admissibility of sorts (Axiom 0):** a sort earns a geometric χ - or T -reading only where a stated map preserves its native combination; most domains do not.
- **Verbs on that unit:** six, set out in the table above, each with the counterpart that takes something back where one exists. Schoolbook $+$ is reserved for **num**.

Later sections distinguish a numeric encoding from a lexical encoding on the same sphere, so that a report may carry words and numbers together without forcing them through $\uplus$.

1.4 Scope of the claim

The actual claim is narrow. Counting ($1 + 1 = 2$) is not the only valid arithmetic for combining units of information. A combination rule should be chosen for what the unit *is* — a count, a color, a direction, a word — not applied by default because it is the arithmetic everyone already knows. The sphere, phase, and the operators $\uplus$, $\oplus$, $\boxplus$, T , offset, and $\bowtie$ are one proposed way to make that claim precise and checkable. They are not themselves the claim, and they are not offered as a general theory of language, physics, biology, or computation. The paper does not attempt to compute the governing equations of those systems. It attempts to arrive at higher-order realities by reading many co-present units of truth — seated units, whether pinned by a lexicon or by a learned map — through a path, by Collapsing them losslessly into one counted atom, or by letting external seats rest under assembly. That is a computation of *presence, phase, order, and rest*, not a substitute for a field equation. The enclosure’s energy is a bookkeeping rule for seats $\vec{P}$; it is not a model of gravity or of matter. The storm of Section 8, and the table and chair, forest and pathogen cancellation of the companion paper on Inference, are costumes for the $\uplus$ -versus- T split: presence versus path. Related-work paragraphs (Section 10) draw boundaries against existing formalisms. None of that is a demonstrated model of weather, ecology, or epidemiology, or a replacement for quantum mechanics, embeddings, or transformers. A later widening of a costume into a domain claim would need its own evidence. It is not an extension of the illustration. Section 2.1 states this limit formally: a sort earns a geometric reading only where a structure-preserving map into Mix or T exists at a stated fidelity, and most physical domains do not offer one.

Every law in this paper is deterministic: the same starting units, read at the same time where any of them carries an expiry (Section 2.11), give the same result, step for step. Where a claim rests on a starting configuration, the configuration itself is printed, as in the worked examples of Sections 4 and 8 and the enclosure of §9.4. How a reader generates configurations of their own — by hand, or by any library under any seed — is left open; the printed one is the shared base. Printing it makes a result replayable. It does not make the result right. A script that recomputes the printed numbers from their printed inputs accompanies the paper as a check; it is not part of the algebra.

Composition. Diverse units come together in one space by composition, not by combination. Bundle keeps their presence and drops repetition; Collapse keeps every vector and every count, whatever the sorts. Neither computes anything across the units, and either can be taken apart again, by Remove or by UnCollapse (Section 3.4). The whole is read as one unit through its derived pair, a reading that forgets while the atom is kept; normalized into $[0, 1]$, that pair may be seated as a unit of the next level (Section 3.1). Mix is different in kind: it combines, and only states that share a direction and a sort — one domain, with one underlying law; across sorts the type table says no or open (Table 2). How differing realities might combine in a higher space is left to Assembly and Inference.

What kind of structure this is. In this paper an arithmetic is a unit, the operations on it, and the properties each operation is stated and checked to have. The unit is the typed point $x = (r, \hat{n}, \chi, \tau)$, with the atoms and bowls built from it. Bundle is idempotent, commutative and associative, and Remove takes a member back. Collapse is commutative and associative, not idempotent, and UnCollapse takes a core back. Mix is commutative, not idempotent, not associative under the cap, partial in sort and in direction, and has no inverse. T is associative, has an identity and an inverse, and is not commutative. Offset copies a walk and Assembly relates two units without fusing them; neither is stated in these terms. Most of these follow at once from a definition: Bundle is set union, Collapse concatenates cores that equality compares as a multiset, Mix’s phasor sum is symmetric, and T composes rotations. The rest are shown where the verb is defined, by proof or by a printed counterexample: Mix’s failure of associativity under the cap, T ’s failure to commute, and the absence of an inverse for Mix (Section 3.2).

What the algebra promises, and what a use supplies. For any use, the algebra promises that its laws are deterministic given the reading time; that a result may be seated again as a unit of the next level; that no source is consumed and every result is a new atom naming its operands by a backward link; that no atom is edited; and that no verb moves a vector from one core to another, so that operating on one core never alters another. A use supplies what the algebra does not decide: the seats, since a lexicon or a learning rule pins the units and the algebra mints neither names nor relations; the reading time; the comparison rule for any grouping by meta vectors; the grain at which atoms are seated, which is the finest piece UnCollapse can take back; the resolution at which measured readings are made comparable; and the named conventions — the clip, $\alpha = \pi r$, the softening length, the shell width and, for the enclosure, η , δ , the census cut and ϵ_{allow} . Three choices in the algebra itself are stipulated, not derived, and are stated as such where they occur: the clip that bounds strength (Section 2.2), the refusal of Mix at differing directions (Section 2.7), and the form of the assembly energy with its softening length (Section 2.9). None of the promises depends on where, how, or by which path or process the algebra is used.

The next section turns these definitions into formal mathematical axioms.

2 Axiomatic Foundations Of Pattern Arithmetic

2.1 Axiom 0: Admissibility of Sorts

The axioms that follow govern how a state $x = (r, \hat{n}, \chi, \tau)$ behaves once it sits on $\mathcal{S}$. They say nothing about which sorts τ have earned the right to sit there with a geometric reading attached. That prior question is this axiom, and it is separate from Table 2 in Section 6.4: the table says which *operators* two already-admitted sorts may use on each other; this axiom says which sorts may claim χ and T mean something about their own domain in the first place.

The test. A sort τ has a domain D_τ and, in the world, its own native way of combining two members of that domain, written $a \circ_\tau b$ for $a, b \in D_\tau$. Call τ *geometrically admissible* if there is a map

$$\varphi_\tau : D_\tau \longrightarrow \mathcal{S}, \quad \varphi_\tau(a \circ_\tau b) \approx \boxplus(\varphi_\tau(a), \varphi_\tau(b)) \quad \text{or} \quad T_{\varphi_\tau(a)}(\varphi_\tau(b)), \quad (1)$$

holding at a stated fidelity (exact, or approximate with the approximation named). Equation (1) is the Tier 1 test. Tier 0 admission, below, does not require it. If no such φ_τ exists, τ is *geometrically inadmissible* for Tier 1: it may still occupy a point on $\mathcal{S}$, but only under the inert treatment described below, never with a claim that χ or T tracks anything the domain itself does.

Three tiers.

- **Tier 0 — default, unstated.** `word` and `num` need no admissibility argument because neither makes a physical claim through the geometry. `num` has an exact, invertible *chart* (Section 6.2): decode, add in $\mathbb{Z}$, encode. That chart is not a Mix or T homomorphism — Mix on it is the clock trap of that same section, which is why Mix on `(num, num)` is type-refused. `word` makes no claim at all — T on a word is bookkeeping (order, presence), never a statement about linguistic physics. Mix applied to two word-states is inert geometry, not a χ -reading of language. Both are silently admitted at this inert level; nothing further needs to be said to use them.
- **Tier 1 — explicit inclusion.** A sort earns a genuine χ -reading only by stating its φ_τ and the fidelity it holds at. Musical pitch class is close to exact: octave equivalence makes pitch native to a circle, so χ for a note is close to literal, not borrowed. Color is close but not unique: additive light mixing is linear in some models and only approximately phasor-like in others, so an inclusion of `color` must say *which* reading it means — Mix under $\boxplus$ and Mix under linear RGB blend are different, non-interchangeable inclusions, and a paper that admits color must pick one and say so. This paper picks phasor Mix on a toy $0^\circ/120^\circ/60^\circ$ chart (Axiom 5). It does not pick CIE, and it does not pick linear RGB. A second Tier-1 sort, `chladni`, is worked in full in Section 2.7, where Equation (1) is met exactly within a mode and shown unavailable across modes.
- **Inadmissible — no φ_τ exists.** Some domains have a native combination law with no structure-preserving map to $(\boxplus, \boxtimes, T, \text{offset}, \boxtimes)$ at any useful fidelity. A snowflake’s native symmetry is the discrete point group D_{6h} , and two ice structures

meeting is a diffusion-limited accretion process governed by temperature and humidity gradients — not a continuous rotation in $\text{SO}(3)$. There is no honest $\varphi_{\text{snowflake}}$ satisfying Equation (1). Such a sort is not barred from the register outright: it may still be loaded as a point, but only demoted to the Tier-0 treatment already given to **word** — T used for order or presence only, no χ -reading granted, and no claim that a rotation of its point corresponds to anything the domain does. What is barred is Tier-1 status: an inadmissible sort may never be handed a stated φ_τ , because none would be true.

Sort	Native law $\circ_\tau$	Mix/ T map?	Tier
num	schoolbook +	no (chart only)	0
word	none claimed	n/a	0
note-pitch	pitch-class sum	near-exact	1
color	additive light	phasor Mix (here)	1
chladni	modal superposition	exact, within a mode	1 [§]
snowflake	D_{6h} growth	no	T1 no; T0 yes

Table 1: Admissibility is a property of the sort, decided before Table 2’s operator legality is even asked. Equation (1) is the Tier 1 test; Tier 0 admission does not use it. A sort absent from this table is undecided, not admitted; treat it as inadmissible until a φ_τ is stated. Collapse is not tested by this table: it preserves each contribution rather than computing a native law, so admissibility for Collapse is a same-sort-versus-cross-sort question, answered in Table 2 instead, not a native-law match asked here. [§]**chladni** is admitted at Tier 1 with a stated restriction: exact for superposition within one mode, undefined across modes (Section 2.7).

This axiom is also what gives Section 1.4 teeth. “Not a general theory of physics, biology, or computation” is not only a promise of restraint; it is a consequence of Equation (1) failing for most physical domains a reader might otherwise expect the sphere to cover.

2.2 Axiom 1: The Normalized Unit Truth State ($r \leq 1$)

A pattern is one typed point, not a field painted over the globe. The working atom is the *information atom*, or *info atom*; the words *unit* and *info atom* are used interchangeably throughout, and *state* names the same object read as a point of $\mathcal{S}$. It is

$$x = (r, \hat{n}, \chi, \tau), \quad \begin{array}{ll} r \in [0, 1], & \hat{n} \in S^2, \\ \chi \in [0, 2\pi), & \tau \in \{\text{word, num, color, note-pitch, chladni, } \dots\}. \end{array} \quad (2)$$

Equivalently, $\hat{n}$ may be written in polar angles (θ, ϕ) , so the geometric part of the info atom is the polar chart (r, θ, ϕ, χ) of the unit, used by the operators below; it leaves out the sort τ , which does not move the arrow. The sort τ says which buttons may be pressed (Section 6). A collection (bowl) is a separate finite set $B = \{x_1, \dots, x_n\}$ of such atoms. The sphere holds one current occupant of the register; the bowl is who is present.

The teaching atom is this 4-tuple: one arrow, $Q = 1$. Axiom 7 allows an atom to hold Q such arrows under one closure, each still capped at $r_i \leq 1$, plus one uncapped derived sum that is not itself an informational vector. That Collapsed structure does not

replace Equation (2). Every informational arrow in the Collapsed atom is an instance of this axiom. Operators that need a single pair $(r, \hat{n})$ read the derived sum as a weight or a direction; as the actor of T in three dimensions, as it is (Axiom 7), and beyond three dimensions not from the derived sum at all, but from $N_{\text{dim}} - 2$ of the Collapsed atom's own orthonormal arrows directly (§4.4). They do not mint a second kind of unit.

The cap $r \leq 1$ applies to *that* arrow. It is a normalized strength / activation coordinate, not a global probability density and not a demand that all arrows on $\mathcal{S}$ integrate to one. Two states may occupy different directions at once; each still carries its own strength $r \leq 1$. The sphere is the stage. The unit is the vector. $r = 1$ is a full-strength mark. Nothing in this axiom sums the globe into a single global budget. A field of many atoms, if wanted later, is a different object (for example a superposition of Dirac masses at those points), not the atom itself.

How the bound is imposed is a choice. This paper bounds by clipping: Mix returns $r' = \min(1, |z'|)$, so $r = 1$ is attained whenever a phasor sum reaches unit length. Clipping is not injective. Every sum of length one or more returns the same r' , and agreement past that point is not recorded (Section 2.6). An order-preserving bijection from $[0, \infty)$ onto $[0, 1)$ — $|z| \mapsto |z|/(1 + |z|)$, say — would bound without discarding, since distinct sums would keep distinct strengths, and it would leave $z = 0 \mapsto 0$ and the black of Axiom 2 untouched.

The clip is kept here, and the trade is stated rather than argued. Under such a bijection $r = 1$ becomes a limit and not a value: the saturated pole of Axiom 3 would be approached and never occupied, and $\alpha = \pi r_w$ would never reach the half-turn of Section 4.3. Two attained poles are worth more to this paper than lossless accumulation, because no construction here sums a large bowl. A setting that does — a verdict accumulated from many units, say — would weigh the same trade the other way, and should revisit this line rather than inherit it.

2.3 Axiom 2: The Absolute Null State ($r = 0$, Black)

The lower bound of a single unit is vanishing strength:

$$r = 0 \quad \implies \quad x = \mathbf{0} \quad (\text{Black}). \quad (3)$$

Direction and phase are then unused: there is no arrow to point. Black is an empty seat, not a field that is zero at every angle. As an atom it holds no arrow at all: one seat and no vector ($N = 1$, $Q = 0$ in the counts of Axiom 7), and a vector of strength zero is this empty seat, not an arrow.

2.4 Axiom 3: Saturation (White)

White is the upper pole of the same bounded stage. It is not an infinite norm. Two readings are allowed; neither leaves the cap $r \leq 1$.

White as a colour. A single arrow may take a full-spectrum name (white light as one identity). That state is still one 4-tuple with $r \leq 1$.

White as a full house. The bowl may be saturated: many directions occupied, many hues present. Collecting further distinct shells fills more seats; it does not raise any arrow past $r = 1$. $\uplus$ does not grow strength. Mix may grow a single r until the cap. The stage looks white because the palette is on. The catalogue of names is large — on the order of a 24-bit palette, ~ 16 million colours per direction — and may be called practically endless. No unit is $r = \infty$.

$$\text{White} = \text{saturation of the bowl, or a full-spectrum name.} \quad (4)$$

Black is empty. White is as full as this construction allows. The menu is huge. The arrows stay finite. Density in this paper means many identities can share the stage without overwrite — many directions, and many phases per direction — each still capped at $r \leq 1$. It does not mean infinite information, and it is not a storage theorem against a voxel grid.

One atom carries a modest, bounded amount before it runs out of room. Two routes exist for more, and they are not the same route. The first is to seat a second atom: $\uplus$ gathers it into the same bowl (Section 2.5), which is how this construction represents anything beyond what one vector alone can hold, and is the route already in use throughout this paper. The second is to raise the ambient dimension, $S^2 \rightarrow S^{N_{\text{dim}}-1}$ (Section 11.1), giving a single atom more room on the sphere it already occupies. The second route is conditional on the first: it relieves crowding among many co-present atoms, and a bowl too small to crowd gains nothing by it — two atoms do not need a millionfold larger stage to stay apart.

2.5 Axiom 4: Bundle and Remove ($\uplus$, $\setminus$)

$\uplus$ *collects* a unit into the bowl. It does not Mix colours and it does not turn an arrow. A bowl is a finite subset of $\mathcal{S}$, and $\uplus$ is union on such subsets:

$$\uplus : 2^{\mathcal{S}} \times 2^{\mathcal{S}} \longrightarrow 2^{\mathcal{S}}, \quad B_1 \uplus B_2 = B_1 \cup B_2.$$

A bare state A appearing in an $\uplus$ -expression denotes its singleton bowl $\{A\}$. With that convention, identical names occupy one seat:

$$A \uplus A = A, \quad \text{that is,} \quad \{A\} \cup \{A\} = \{A\}, \quad (5)$$

while distinct names take separate seats, $A \uplus B = \{A, B\}$, so opposite marks may sit together: $\{+1\} \uplus \{-1\} = \{+1, -1\}$. Idempotence, commutativity and associativity of $\uplus$ are those of union, and none of the three needs separate postulation. Bundle is sort-blind: any two seated units may share a bowl, whatever their sorts, since a bowl records presence and presence makes no claim about a domain (Table 2). Sort discipline governs what may later be *done* with the bowl, not who may sit in it. The bowl is a set of states, not a blended 4-tuple.

$\setminus$ *removes* a unit, $\setminus : 2^{\mathcal{S}} \times \mathcal{S} \rightarrow 2^{\mathcal{S}}$. Removing the last occupant leaves the empty bowl:

$$\{A\} \setminus A = \emptyset. \quad (6)$$

The empty bowl $\emptyset$ and the null state $\mathbf{0}$ of Axiom 2 are distinct objects: $\emptyset$ is a bowl with no occupant, $\mathbf{0}$ is an occupant with $r = 0$. Only $\mathbf{0}$ is a state, which is why $T_{\mathbf{0}}$ is

defined (Section 4.3) and $T_\emptyset$ is not. $\setminus$ undoes a Bundle. It is not anti-phase interference. Idempotence then gives the counting-trap

$$A \uplus A \uplus A \uplus A \setminus A = \emptyset \quad (\text{the empty bowl}). \quad (7)$$

Four gathers of the same name are still one seat; one removal empties it. Schoolbook $1 + 1 + 1 + 1 - 1 = 3$ lives only in **num** mode.

2.6 Axiom 5: Mix as a reading ($\boxplus$)

A blended 4-tuple appears only when Mix *reads* the bowl. Write $\text{Mix} = \boxplus$. Mix is a *phasor* operator: it combines χ among states that already share a direction $\hat{n}$. Direction belongs to T and to the rotations of §2.8; Mix never moves an arrow onto a new axis. Mix of states with differing $\hat{n}$ is refused (Section 2.7). The output is $(r', \theta', \phi', \chi')$ with $r' \leq 1$, or black. T_w is a one-argument rotation and phase-shift of an occupant, parameterized by an actor w . Mix is a combination of peer states. It is not a member of the group T_w lives in, and it does not compose with T_w the way two ordinary T 's do. The two give different answers on the same pair: for unit red at $\chi = 0^\circ$ and unit green at $\chi = 120^\circ$ sharing a direction, $T_{\text{green}}(\text{red})$ shifts phase additively to $\chi' = 120^\circ$, while $\text{red} \boxplus \text{green}$ takes the phasor resultant, $\chi' = 60^\circ$. No choice of actor makes T return 60° here, because T adds phase where Mix sums phasors. Three further properties separate them: T_w is asymmetric in actor and patient while $\boxplus$ is commutative; T_w leaves r invariant while $\boxplus$ may grow it to the cap (Axiom 3); and every T_w has an inverse w^* while $\boxplus$ has none.

Which sorts may read a Mix result as a claim about their own domain is settled by Axiom 0, not here: Tier 1 for **color** and **note-pitch**, inert geometry for **word**, type-refused for **num**.

Two states at a common direction. Treat strength and phase as a phasor $z = re^{i\chi}$. A domain reading requires A and B to share a sort (§2.1 above), so the sort carries through unchanged:

$$z' = r_A e^{i\chi_A} + r_B e^{i\chi_B}, \quad r' = \min(1, |z'|), \quad \chi' = \arg(z'), \quad \tau' = \tau_A = \tau_B, \quad (8)$$

and $(\theta', \phi') = (\theta_A, \phi_A)$. If $z' = 0$, the result is black. Equal unit paints at 0° and 120° give $z' = e^{i\pi/3}$, so

$$(1 \angle 0^\circ) \boxplus (1 \angle 120^\circ) = (1, \theta, \phi, 60^\circ, \text{color}) \quad (\text{yellow}).$$

This paper's color inclusion (Axiom 0, Tier 1) is that phasor Mix on a toy $0^\circ/120^\circ/60^\circ$ chart. It is not CIE, and it is not linear RGB. The unequal pair $r = 0.6$ at 0° and $r = 0.8$ at 120° gives $z' = 0.20 + 0.6928i$, hence

$$r' \approx 0.721, \quad \chi' \approx 73.9^\circ.$$

Equal anti-phase marks, same place, give $z' = 0$ (black). A state mixed with itself at the same phase is not idempotent: $A \boxplus A$ gives $z' = 2re^{i\chi}$, so $r' = \min(1, 2r)$. Strength grows until the cap clips it. That is distinct from $A \uplus A = A$.

Reading a whole bowl. Mix over a bowl B whose members share a direction $\hat{n}$ and, for a domain reading, a single sort τ is defined in one step, not as a fold:

$$\text{Mix}(B) = (\min(1, |z_B|), \hat{n}, \arg z_B, \tau), \quad z_B = \sum_{x \in B} r_x e^{i\chi_x}. \quad (9)$$

The binary $\boxplus$ of Equation (8) is the case $|B| = 2$. The one-step reading is written $\text{Mix}(B)$, not $\boxplus(B)$, so that it is not read as a fold of the binary operator. The one-step form is necessary, not stylistic, because the cap makes the binary operator non-associative: with $A = 1\angle 0^\circ$, $B = 1\angle 0^\circ$, $C = 1\angle 180^\circ$,

$$(A \boxplus B) \boxplus C = \text{black}, \quad A \boxplus (B \boxplus C) = 1\angle 0^\circ,$$

since $A \boxplus B$ is clipped from $2\angle 0^\circ$ down to $1\angle 0^\circ$, and the strength discarded by that clip is exactly what C would otherwise have cancelled. Equation (9) on the same three states sums first and caps once:

$$z_B = e^{i0} + e^{i0} + e^{i\pi} = 1, \quad \text{Mix}(\{A, B, C\}) = (1, \hat{n}, 0^\circ, \tau),$$

one reading, reached without choosing an order. $\boxplus$ is commutative, and Equation (9) is order-independent, so a bowl — which carries no order — has exactly one Mix reading. Iterated binary Mix does not, and is not what Equation (9) means. Non-associativity is the price of the cap of [Axiom 1](#): without $\min(1, \cdot)$ the phasor sum would be ordinary complex addition and associativity would be free.

Reading a Collapsed atom. Mix reads a Collapsed atom A the same way, with one difference: the sum runs over every informational vector it holds, so a repeated core adds again, $z_A = \sum_{i=1}^Q r_i e^{i\chi_i}$, and $\text{Mix}(A)$ is Equation (9) with z_A in place of z_B . The set bowl of the same atom counts a repeated member once, so the two readings agree exactly when nothing repeats ($D = N$), as Bundle and Collapse do. Two copies of $0.3\angle 0^\circ$ read 0.6 as $A \oplus A$ and 0.3 as $A \uplus A$. The conditions are unchanged: one direction and, for a domain reading, one sort, and no reading across them.

$\uplus$ does not run this rule. Two shells in a bowl have no resultant angle until Mix (or a T -path) is applied. A reading returns a state and leaves the bowl intact: $\text{Mix}(B)$ does not consume B , and a T -path draws actors from a bowl without removing them. Only $\backslash$ takes a member out. That is what makes one bowl readable more than once — the same B may yield a Mix result and, separately, a walked path. Bundle is lossless and is undone by $\backslash$; Collapse is also lossless and is undone by $\ominus$, one core at a time; Mix is lossy, since it discards both order and any strength above the cap. That is why $\backslash$ inverts $\uplus$, $\ominus$ inverts $\oplus$, w^* inverts T_w , and nothing inverts $\boxplus$ — named UnMix , $\boxminus$, in Section 3.2.

Saturation and levels. A sum carries no bound of its own; the cap of [Axiom 1](#) supplies one. A bowl of many agreeing units therefore reaches $r' = 1$ quickly, and further agreement past that point is not recorded. This is worth reading as a remark about levels rather than a defect in the cap. A phasor sum is a reading of units that stand at one level together, and repeated samples of a single unit do not stand at one level; a bowl that saturates early is often a level that has been flattened (Section 3.1). Strength does not accumulate across levels: a reading returns a state whose r is legal by construction — $\min(1, |z_B|)$ for Mix, and the occupant's own r for a path — and that

state is seated afresh at the level above. Composition hides the question, since a path over many actors stays in $SO(3)$ and never saturates, so it is Mix rather than the walk that makes a flattened level visible. What the cap does not supply is a way to average, which needs the count that $\oplus$ discards (Section 2.5); reducing repeated samples to one seated unit is therefore work for the level below, not for $\boxplus$.

2.7 A worked Tier-1 inclusion: `chladni`

Color is admitted on a declared toy chart. A second sort is admitted here on stronger ground, to show that Equation (1) can be met exactly rather than approximately, and by physics unrelated to either color or language.

Native law. A thin plate driven at one frequency responds as a superposition of its eigenmodes $\varphi_j(x, y)$, each carrying a complex amplitude $A_j = a_j e^{i\chi_j}$; the visible figure is the nodal set of $w = \sum_j A_j \varphi_j$. So $\circ_{\text{chladni}}$ is complex addition of mode amplitudes. Two drives on the same mode add as complex numbers.

Chart. One state is one mode. Let $\hat{n}$ carry the mode index under any injective map $(m, n) \mapsto (\theta, \phi)$, let χ be that mode's phase, and let $r = a/a_{\max}$ be its amplitude normalized to a declared saturation $a_{\max}$.

The map, and where it is exact. Within one mode, $\hat{n}$ is fixed, and

$$\varphi_{\text{chladni}}(A \circ B) = \varphi_{\text{chladni}}(A) \boxplus \varphi_{\text{chladni}}(B) \quad (10)$$

holds identically while $|z| \leq 1$, because complex addition of amplitudes and phasor addition of (r, χ) are the same operation. Nothing is fitted. Note also that the chart's arbitrariness costs nothing here: same-mode combination never moves $\hat{n}$, so Equation (10) places no constraint on the mode-to-direction map. Two consequences are physical and immediate. Equal amplitudes in anti-phase give $z = 0$: the mode is driven to silence and its figure disappears, which is Equation (8) returning black. Two drives at 0° and 120° give $r = 1$ and $\chi' = 60^\circ$ — the same arithmetic as red $\boxplus$ green = yellow, on unrelated physics.

The restriction, which is physical. Across modes the map is not available, and not for want of effort. Mode space is not closed under addition: a sum of two distinct modes is not a mode, so it cannot be seated as a single state under any chart. Cross-mode coexistence is therefore $\oplus$ — a bowl of modes — and the resulting figure is a reading of that bowl which is not $\boxplus$. This is the question of Section 2.7 met in a concrete domain, and it is the evidence that question is closed on: refused, not left waiting.

Fidelity claimed, and three limits. Exact in phase and amplitude below saturation, within one mode; undefined across modes. Three limits are stated rather than argued away. The cap $r \leq 1$ is a normalization to a declared $a_{\max}$, not a fact of linear plate theory, so Equation (10) clips above saturation, exactly as the color inclusion does. Direction is a label here and nothing more, so no meaning is claimed for T on a `chladni` state: rotating a mode index is not an operation the domain performs. And the figures most often reproduced are degenerate-mode combinations, which fall on the cross-mode side; what is admitted exactly is the simpler case, not the famous one.

Mix across differing directions: refused. [Axiom 5](#) defines Mix only among states that share a direction $\hat{n}$, where it is phasor combination on χ . This paper states directly what was once left open: Mix refuses states at differing directions, making the operator partial in direction exactly as it is already partial in sort (Table 2). An earlier draft carried a provisional recipe — normalize the vector sum $r_A\hat{n}_A + r_B\hat{n}_B$ — and withdrew it, because it was the only place in the construction where $\hat{n}$ would move by something other than a rotation. Everywhere else, direction is the exclusive property of $\text{SO}(3)$ (§2.8) and of T , and a blend that slides an arrow between axes was never a rotation. That withdrawal is the reason for the refusal, not merely a precedent for it.

The `chladni` inclusion supplies the direct evidence: cross-mode coexistence — states at different directions within one admitted sort — is $\uplus$, a bowl, and the figure read from that bowl is not $\boxplus$ (§2.7). Mode space fails to close under addition for a stated physical reason, not a gap in this paper’s own construction, and nothing in this paper’s machinery supplies what physics itself does not: a map from two directions to a third that is anything other than a rotation. Of the three routes once held open, refusal is the one adopted here, and it is a claim about $\boxplus$ at one shared direction, not a claim that states at differing directions cannot be combined. Rotating one state onto the other’s axis first is available as T followed by Mix whenever a use names the axis; what Mix cannot do is choose it. A genuine two-stage operator — lift both states to a higher space, where they sit side by side without interfering, and read them there — is neither adopted nor barred: a bowl and a Collapsed atom already are such a lift, and V_{derived} of [Axiom 7](#) is one reading of it, taken in the ambient space. A reading that returned a single state at a stated direction would be a new operator, with its own stated map and its own place in the type table, under the same “undecided, not admitted” rule as [Axiom 0](#). The `chladni` case supports refusal at one shared direction: a concrete domain in which the operation is undefined.

Refusal costs nothing already needed. Two same-sort states at differing directions remain fully available to the rest of the algebra: $\uplus$ keeps them as separate, order-blind seats, exactly as `chladni`’s own bowl of modes does; $\oplus$ keeps them losslessly as two vectors inside one Collapsed atom, since Collapse requires neither a shared direction nor a shared sort ([Process principle: Collapse](#)). What is refused is only the one further claim that was never earned: reading two different directions as if they already answered one shared question.

2.8 Rotation keeps $\hat{n}$ on the sphere

A rotation $\mathbf{R} \in \text{SO}(3)$ turns the direction $\hat{n}$ and leaves strength, phase, and sort alone:

$$\mathbf{R} \cdot (r, \hat{n}, \chi, \tau) = (r, \mathbf{R}\hat{n}, \chi, \tau), \quad \|\mathbf{R}\hat{n}\| = \|\hat{n}\| = 1. \quad (11)$$

$\hat{n}$ stays on S^2 because a rotation preserves norm, which is a fact about rotations. That r, χ and τ are left alone is the action chosen here, so the unit is preserved by choice and not by theorem. This is the geometric engine of T , not a second blend.

2.9 Axiom 6: Assembly ($\bowtie$)

Bundle lists who is present; Mix fuses peers into one state; T walks an ordered path. A fourth question remains: given two units, how strongly do they belong together, and where do they touch? Assembly, written $\bowtie$, answers it by adding a relation while leaving

both units intact — it is neither $\uplus$ (which states presence and no relation) nor $\boxplus$ (which fuses and loses both identities):

$$A \bowtie B = (A, B, \kappa_{AB}, U_{AB}). \quad (12)$$

Both operands survive unchanged. What is new is the pair (κ_{AB}, U_{AB}) : a contact point and an interaction energy.

The interaction energy.

$$U_{AB} = -\frac{r_A r_B}{2} \cdot \frac{3 + \hat{n}_A \cdot \hat{n}_B}{(d_{AB}^2 + \varepsilon_{AB}^2)^{3/2}}, \quad \varepsilon_{AB} = r_A + r_B. \quad (13)$$

Three factors, each answering a different question. The product $r_A r_B$ asks how strong the two units are; a weak partner lowers the whole. The dot product $\hat{n}_A \cdot \hat{n}_B$ asks how well their directions agree, and is bounded: the bracket ranges over $[2, 4]$ only, so alignment can shift the result by at most a factor of two. The denominator asks how far apart they sit, which requires the enclosure of Section 9 and is the one term that reaches outside the bounded stage of Axiom 1. The softening length $\varepsilon_{AB} = r_A + r_B$ is a convention, not a consequence of the axioms: reading r as a radius, as the volume of Axiom 7 does, two units cannot approach closer than the sum of their own extents.

The contact point.

$$\kappa_{AB} = \hat{n}_A + \hat{n}_B, \quad (14)$$

the raw, unnormalized sum. Leaving it unnormalized is deliberate. The normalized bisector $\kappa_{AB}/|\kappa_{AB}|$ is undefined when $\hat{n}_B = -\hat{n}_A$, and that degeneracy drove a long search for a special case that turned out not to be needed: U_{AB} never required an axis at all, and κ_{AB} is a plain vector sum, defined at every angle. At exactly antipodal orientation it is $\mathbf{0}$ — a computed answer, not a failure, and the same absence of a favoured direction that Axiom 2 reports as black. Its magnitude is a free confidence signal, falling smoothly from 2 (identical directions) to 0 (opposed).

Assembly never repels. For two units of nonzero strength considered alone, $U_{AB} < 0$ always: there is no orientation, no pair of strengths, and no separation at which isolated assembly pushes rather than pulls. This is not an assumption but a property of Equation (13), whose bracket cannot fall below 2. The one boundary is a black unit (Axiom 2): at $r_A = 0$ the energy is exactly zero, no relation being formed with something absent, which is what black already means elsewhere. This is taken as the definition, $U_{AB} = 0$ whenever $r_A r_B = 0$, because the formula alone gives $0/0$ for two black units on one seat. Whether a structure of three or more units can hold members apart is a question for the enclosure, not for the pairwise verb.

Polarity. The difference $|r_A - r_B|$ distinguishes a balanced bond from a lopsided one, and so supplies a criterion for reading $A \bowtie B$ as mutual or as actor-and-patient in the manner of T . It is a label on the relation, not a difference in what the relation does: Equation (13) is symmetric under exchange of A and B , and $U_{AB} = U_{BA}$ exactly.

Where assembly sits among the verbs. Assembly is the first operator in this paper that requires anything outside the 4-tuple. Bundle, Remove, Mix, rotation, and offset all take units on the sphere and return units or states on the same sphere; none of them asks *where* a unit is, because nothing in $(r, \hat{n}, \chi, \tau)$ answers that. Assembly asks, and so is stated here as a verb but completed in Section 9, which supplies the seat $\vec{P}$ (a meta vector, Section 2.11) that d_{AB} measures between. Read without an enclosure, Equation (13) still gives a well-defined ordering of which pairs favour each other most strongly at equal separation.

2.10 Axiom 7: Informational vectors and derived per-atom metrics

A map of this axiom. Each quantity below is defined in the text, in this order, and none enters a verb unless it is named.

Quantity	What it answers	Range
vector $v_i = (r_i, \hat{n}_i, \chi_i)$, sort τ_i	One informational reading, an instance of Axiom 1	$0 < r_i \leq 1$
core; N, Q, D	Cores, vectors, distinct cores; each vector is in one core, and the grouping is part of the atom	$Q \geq 0, D \leq N$
$V_{\text{derived}} = \sum r_i \hat{n}_i$	The raw sum of the vectors; a meta vector, uncapped and not normalized	any length
$\mathcal{I}_{\text{str}}$	How much strength in total	≥ 0
$\mathcal{I}_{\text{coh}} = V_{\text{derived}} /\mathcal{I}_{\text{str}}$	How far the vectors reinforce one another	$[0, 1]$
$\mathcal{I}_{\text{vol}}, \mathcal{I}_{\text{mean}}$	Total and per-vector volume (r^3)	$< \mathcal{I}_{\text{str}}$ if some $r_i < 1$
$\mathcal{I}_{\text{en}}$	Squared magnitude if all vectors were orthogonal	≥ 0
$\mathcal{I}_{\text{agr}}$	Whether the directions agree or cancel (weighted mean cosine)	$[-1, 1]$
$\mathcal{I}_{\text{vcon}}$	How unevenly volume is shared among vectors	$[0, 1]$
$\mathcal{I}_{\text{scon}}(w)$	Whether many vectors share one strength bin of width w	$[0, 1]$
$Q_{\text{eff}} = \mathcal{I}_{\text{str}}^2/\mathcal{I}_{\text{en}}$	Equally strong vectors that carry the same energy	$[1, Q]$

The teaching atom of [Axiom 1](#) is one unit ($N = Q = 1$), one state among several. An atom may also be *Collapsed*: Q informational vectors $v_i = (r_i, \hat{n}_i, \chi_i)$, each carrying its own phase-color $\chi_i \in [0, 2\pi)$ alongside $(r_i, \hat{n}_i)$ and its own sort τ_i — v_i together with τ_i forming, per vector, the same unit [Axiom 1](#) states, still capped at $r_i \leq 1$, and positive: a vector of strength zero is black ([Axiom 2](#)), not an arrow. The vectors of one atom may share a sort or differ in it. They are held in groups; call each group a *core*. A vector is held by exactly one core, as a nucleus holds its electrons, and no verb moves a vector from one core to another. Which core holds which vector is part of the atom, recorded as a containment vector: for each vector, the position of its core in the atom's list of cores. N is the number of cores and Q the number of vectors. The two counts are independent: a core holds any number of vectors, one, many, or none, so $Q \geq N$ is the usual case but not a rule, $N = Q = 1$ is the teaching atom, and $N = 1, Q = 0$ is the empty atom, which is black. An atom reaches this form by Collapse (below) or by being seated in it directly, by whatever process supplies the seat — a lexicon or a learning rule, as for any unit; this paper constructs neither. This is not a second species of unit and

not a field on the globe. It is several teaching atoms held under one closure — distinct phase-colours among them coexist without being forced through Mix, the Collapsed atom’s own instance of two identities occupying one direction without overwriting one another (Axiom 1). One further vector, computed from them, may sit alongside the vectors of each core and alongside those of the atom as a whole, the first of the meta vectors of Section 2.11; for the atom,

$$V_{\text{derived}} = \sum_{i=1}^Q r_i \hat{n}_i,$$

the raw, unnormalized sum of the informational vectors’ geometric part, computed after and from them, never including itself; over one core’s vectors the same sum gives that core’s, and the atom’s is the sum of its cores’; χ_i does not enter this sum, the same way τ does not participate in T ’s rotation; nor do τ_i or the core a vector sits in, so the sum runs over every vector of the atom, whatever its sort — χ_i is available wherever a per-vector reading is named, silent in the seven metrics below, none of which is stated in terms of it. What one sort’s vectors, or one core’s, contribute is read from the sub-atom that holds them (Section 3.4); a sub-atom is another atom, with a V_{derived} of its own. V_{derived} is uncapped and excluded from both counts, N and Q : it needs no τ -justified admissibility of its own (Axiom 0), the same status the companion paper² already grants its own uncapped witness-sums. Its magnitude and direction read as an effective $(r, \hat{n})$ pair — $r = |V_{\text{derived}}|$, $\hat{n} = V_{\text{derived}}/r$ — standing in for the atom wherever a weight or a direction is required: Assembly’s $r_A r_B$ (Axiom 6), a camp body’s alignment — neither involves a rotation or $\mathcal{S}$ -membership. It may act as the actor w of T (§4.2) as it is: a meta vector read as the pair $(|V_{\text{derived}}|, V_{\text{derived}}/|V_{\text{derived}}|)$, uncapped and not normalized, since it is not an informational vector and is not treated as one. A strength above 1 is then an angle above 180° , taken modulo 360° : three vectors at $r = 0.5$, perfectly aligned, give $|V_{\text{derived}}| = 1.5$ and a turn of 270° . Because $|V_{\text{derived}}|$ grows with the number of vectors, the turn follows size, not content: $|V_{\text{derived}}| = 51.877$ and 5187.748 turn by 337.9° and 314.6° (the sizes of a hundred and ten thousand identical colours, below). A use that wants a turn that follows content takes $\mathcal{I}_{\text{coh}} = |V_{\text{derived}}|/\mathcal{I}_{\text{str}}$ (defined below) as the strength, a choice at the call site. Every worked T path in this paper takes a teaching atom, and a Collapsed-atom-as-actor example is not claimed here. A bowl is not an actor either: it is a collection, and T takes one unit as its actor. For $Q = 1$, V_{derived} coincides with the atom’s own vector, already in $\mathcal{S}$, and nothing changes.

Where V_{derived} places a colour, it is read by direction, not by phase and not by length: the colour lives in $\hat{n}$, and χ stays for mixing. Length is not colour, since it grows with the count; one, a hundred and ten thousand identical colours give 0.519, 51.877 and 5187.748. Direction alone is many-to-one: fifty red with fifty green, and one hundred uniform yellow, share the direction $(0.707, 0.707, 0)$, and $\mathcal{I}_{\text{coh}}$ tells them apart, 0.707 for the blend and 1.000 for the yellow. A colour’s locator is therefore the direction together with $\mathcal{I}_{\text{coh}}$.

A bowl read as its members’ vectors holds these quantities by the same formulas, counting a repeated member once, and a Collapsed atom counts it every time; meta vectors stay with atoms. Seven scalar metrics follow from the same Q informational vectors, all of them, whichever core holds each, among the cores valid at the reading

²*Pattern Arithmetic: Inference from Bowls and Paths*, which builds inference machinery — reference atoms, classification, this same τ -exempt status for uncapped witness-sums — on top of the atoms and operators defined here.

time (Section 2.11), each answering a different question, none used by this Axiom’s own Assembly (§2.9) or by Motion (Section 9) unless named. All seven are written $\mathcal{I}$, for the information atom, with a subscript that names the metric:

$$\mathcal{I}_{\text{str}} = \sum_{i=1}^Q r_i, \quad \mathcal{I}_{\text{vol}} = \sum_{i=1}^Q r_i^3, \quad \mathcal{I}_{\text{mean}} = \frac{\mathcal{I}_{\text{vol}}}{Q}.$$

$\mathcal{I}_{\text{str}}$, atom strength, is the correct denominator for atom-level coherence, $\mathcal{I}_{\text{coh}}$, the ratio $|V_{\text{derived}}|/\mathcal{I}_{\text{str}}$, structurally identical to the companion paper’s own mean resultant length one level down — reading 1.0 for perfectly aligned informational vectors and falling smoothly toward 0 as they scatter. $\mathcal{I}_{\text{vol}}$, atom volume, is strictly smaller than $\mathcal{I}_{\text{str}}$ whenever any $r_i < 1$, a forced consequence of the $r \leq 1$ cap rather than an assumption: cubing a number below 1 always shrinks it, exactly as treating r as a literal radius and r^3 as the volume it encloses would predict. $\mathcal{I}_{\text{mean}}$, mean vector volume, is the average per-vector share of that total. None of the three participates in Motion; atom volume, in particular, is deliberately excluded from it and exists here purely as a descriptive metric of the atom. The names “mass,” “density,” and “atom energy” are deliberately not used for $\mathcal{I}_{\text{vol}}$, $\mathcal{I}_{\text{mean}}$, and $\mathcal{I}_{\text{str}}$ respectively: the first two because neither participates in momentum, kinetic energy, or any calculation the word “mass” would ordinarily license, and the third because a genuine notion of energy exists below and is not $\mathcal{I}_{\text{str}}$.

A fourth quantity is not derived from strength by squaring it, but stands on its own, the way variance is defined directly rather than as squared standard deviation:

$$\mathcal{I}_{\text{en}} = \sum_{i=1}^Q r_i^2. \quad (15)$$

$\mathcal{I}_{\text{en}}$, the atom’s *information energy*, is the combined squared magnitude the atom’s informational vectors would carry if every pair were mutually orthogonal — confirmed by direct construction, computing $\mathcal{I}_{\text{en}}$ from a set of strengths matches $|V_{\text{derived}}|^2$ computed from those same strengths placed at mutually orthogonal directions, exactly. This is forced, not chosen: an exact identity places $\mathcal{I}_{\text{en}}$ inside the derived vector’s own squared magnitude,

$$|V_{\text{derived}}|^2 = \mathcal{I}_{\text{en}} + 2 \sum_{i < j} r_i r_j (\hat{n}_i \cdot \hat{n}_j),$$

verified to six decimal places. The structure matches the variance of a sum, $\text{Var}(\sum X_i) = \sum \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$, term for term: r_i^2 stands where $\text{Var}(X_i)$ stands, each vector’s own self-contained contribution; $r_i r_j (\hat{n}_i \cdot \hat{n}_j)$ stands where $\text{Cov}(X_i, X_j)$ stands, how two vectors relate. $\mathcal{I}_{\text{en}}$ is the part that remains when every cross term vanishes — the atom’s information energy, prior to any question of how its own vectors align. This is distinct from the companion paper’s own bowl energy (a sum across many separate witnesses): bowl energy sums across atoms, $\mathcal{I}_{\text{en}}$ sums squares within one.

A fifth metric answers a question the other four cannot. $\mathcal{I}_{\text{coh}}$ above is a ratio of magnitudes and so cannot fall below zero: it reads 0 both for informational vectors that merely fail to reinforce and for vectors that actively cancel. The strength-weighted mean cosine similarity across all internal pairs, which we call *atom agreement*, separates these:

$$\mathcal{I}_{\text{agr}} = \frac{\sum_{i < j} r_i r_j (\hat{n}_i \cdot \hat{n}_j)}{\sum_{i < j} r_i r_j}. \quad (16)$$

Each $\hat{n}_i \cdot \hat{n}_j$ is already a cosine similarity, since informational vectors are unit-directed; the denominator makes the aggregate comparable across atoms of differing Q and differing strengths, which the unnormalized sum is not — two atoms with comparably tight internal arrangement but $Q = 2$ and $Q = 5$ show how little the raw sum alone says: $r = 0.85$ throughout, pairwise directions 5.7° apart, gives a raw sum of $+0.72$ at $Q = 2$ (one pair) against $+7.14$ at $Q = 5$ (a five-fold symmetric cone of half-angle 5.7° , ten pairs) — a tenfold spread driven purely by pair count, not by any real difference in agreement, which $\mathcal{I}_{\text{agr}}$ correctly reports as 0.995 and 0.988, nearly identical. $\mathcal{I}_{\text{agr}} = +1$ exactly when all informational vectors align; it is negative when they genuinely oppose, reading $-1/3$ for two antipodal pairs on orthogonal axes (four unit vectors, six pairs, two at cosine -1 and four at cosine 0). Its floor is not -1 in general: three or more vectors in three dimensions cannot be mutually opposed, so achievable disagreement tightens as Q grows while achievable agreement does not — the unnormalized sum has maximum $\sum_{i < j} r_i r_j$, growing as Q^2 , against a hard minimum of $-\frac{1}{2}\mathcal{I}_{\text{en}}$, growing only as Q , which follows directly from $|V_{\text{derived}}|^2 = \mathcal{I}_{\text{en}} + 2 \sum_{i < j} r_i r_j (\hat{n}_i \cdot \hat{n}_j) \geq 0$. Agreement compounds; disagreement saturates. Atom agreement, $\mathcal{I}_{\text{agr}}$, is read alongside $\mathcal{I}_{\text{coh}}$, not in place of it: the first asks whether the vectors agree, the second how much of their strength survives into one resultant.

A sixth metric, *volume concentration*, does for magnitude what $\mathcal{I}_{\text{agr}}$ does for direction. $\mathcal{I}_{\text{str}}$ and $\mathcal{I}_{\text{vol}}$ both report a total and are silent on how that total is spread: two atoms can carry near-identical strength — 2.200 for four vectors of 0.55 each, 2.150 for four of 0.95, 0.95, 0.15 and 0.10 — while one holds it evenly across its vectors and the other concentrates most of it in two. Writing $s_i = r_i^3 / \mathcal{I}_{\text{vol}}$ for the share of volume held by vector i :

$$\mathcal{I}_{\text{vcon}} = \frac{Q^2}{Q-1} \cdot \text{Var}(s_1, \dots, s_Q), \quad Q \geq 2, \quad (17)$$

with $\mathcal{I}_{\text{vcon}} = 0$ by convention at $Q = 1$. $\mathcal{I}_{\text{vcon}} = 0$ exactly when volume is spread evenly and 1 when a single vector holds all of it, at every Q — verified at $Q = 2, 3, 4, 6, 8$ for both extremes. The normalization is what earns this form over the simpler candidates: the largest vector’s share and the Herfindahl index were both tested and rejected, since each reads a perfectly even atom as $1/Q$ and so slides from 0.5 to 0.125 as Q grows from 2 to 8, confounding evenness with size. $\mathcal{I}_{\text{vol}}$ rather than $\mathcal{I}_{\text{str}}$ supplies the shares deliberately: cubing amplifies concentration, so a single dominant vector at $r = 0.95$ against three at $r = 0.2$ registers a share of 0.973, where linear strength would report the same atom as unremarkable.

A seventh metric, *shell concentration*, answers a question none of the first six can: not how strength or volume is shared among distinct vectors, but whether many vectors independently arrive at nearly the same magnitude. $\mathcal{I}_{\text{vcon}}$ is blind to this by construction — four hundred vectors at an identical $r = 0.65$ each receive their own tiny individual share and read $\mathcal{I}_{\text{vcon}} = 0$, while four hundred spread evenly over $(0.2, 1)$ read only 0.0021. A clump is not seen: four hundred strengths spread evenly over $(0.66, 0.69)$ read $\mathcal{I}_{\text{vcon}} = 0.000004$, *more* even than the uniform spread, though every one of them sits in the same shell. Fixing this needs a shell width w , a named, configurable parameter rather than a derived one, defaulting to $w = 0.1$ — ten shells spanning $[0, 1]$. A shell here is a strength bin of width w , not a unit in a bowl. Binning the Q informational vectors by $\lfloor r_i/w \rfloor$, computed in exact arithmetic so that a strength of exactly kw lies in shell k (in floating point $\lfloor 0.3/0.1 \rfloor$ is 2, not 3), into K occupied shells with populations $n_1, \dots, n_K$

and $p_k = n_k/Q$:

$$\mathcal{I}_{\text{scon}}(w) = \begin{cases} 1, & K = 1 \\ \frac{K^2}{K-1} \cdot \text{Var}(p_1, \dots, p_K), & K \geq 2. \end{cases} \quad (18)$$

The $K = 1$ case is stated separately because the general form has a removable singularity there: every vector sharing one shell is maximal concentration, not the zero the formula would otherwise return. $\mathcal{I}_{\text{scon}} = 0$ when occupied shells are evenly populated, at every K ; $\mathcal{I}_{\text{scon}} = 1$ when every vector shares one shell. Checked at $w = 0.1$ and $w = 0.05$: well separated shells (as in 10@0.98, 3@0.3, 400@0.65, giving $\mathcal{I}_{\text{scon}} = 0.908$ at either width) are stable to the choice, while genuinely uniform populations are not: the four hundred strengths spread evenly over $(0.2, 1)$ occupy 8 shells at $w = 0.1$ and 16 at $w = 0.05$, though the concentration score stays 0 at both. w should be reported alongside $\mathcal{I}_{\text{scon}}$ whenever it is used.

Effective vector count. Not an eighth metric: a re-expression of $\mathcal{I}_{\text{str}}$ and $\mathcal{I}_{\text{en}}$, as $\mathcal{I}_{\text{coh}}$ is a ratio of $|V_{\text{derived}}|$ and $\mathcal{I}_{\text{str}}$. Write

$$Q_{\text{eff}} = \frac{\mathcal{I}_{\text{str}}^2}{\mathcal{I}_{\text{en}}}, \quad \text{so} \quad \mathcal{I}_{\text{en}} = \frac{\mathcal{I}_{\text{str}}^2}{Q_{\text{eff}}}.$$

It counts the equally strong vectors that would carry the same energy: $Q_{\text{eff}} = Q$ when every strength is equal, and it falls toward 1 as a single vector dominates, so $1 \leq Q_{\text{eff}} \leq Q$ whenever $\mathcal{I}_{\text{str}} > 0$. It is scale-free — halving every r_i leaves it unchanged — and the largest vector holds at least $1/Q_{\text{eff}}$ of the total strength. It is a *count*, not an evenness score: a perfectly even atom reads $Q_{\text{eff}} = Q$, growing with size, which is the confounding that rejected the Herfindahl index above when used as evenness; as a count that growth is the point, and Q_{eff}/Q is the size-free form. The relation to $\mathcal{I}_{\text{vcon}}$ is exact: with $H = \sum_i s_i^2$, $\mathcal{I}_{\text{vcon}} = (QH - 1)/(Q - 1)$, the Herfindahl index of the cubed shares normalized so that an even atom reads 0; Q_{eff} is the same construction on plain strengths, left as a count. Like the seven metrics it is descriptive and enters no operator.

The empty atom. At $Q = 0$ — black, $N = 1$ — $\mathcal{I}_{\text{str}}$, $\mathcal{I}_{\text{vol}}$, $\mathcal{I}_{\text{en}}$ and V_{derived} are empty sums and equal 0. Two quantities have no value to report and are set by convention, extending the stated convention at $Q = 1$: $\mathcal{I}_{\text{mean}} = 0$ and $\mathcal{I}_{\text{vcon}} = 0$. Four are undefined for the empty atom: $\mathcal{I}_{\text{agr}}$, which needs at least two vectors, $\mathcal{I}_{\text{scon}}$, which needs at least one, and $\mathcal{I}_{\text{coh}}$ and Q_{eff} , which need nonzero strength.

2.11 Meta vectors: what an atom carries beside its content

An atom's informational vectors are what it says: each is capped at $r \leq 1$ and counted in Q . Beside them an atom may carry *meta vectors*: values that describe the atom and are no part of what it says. A meta vector is not capped, is not an informational vector, and is counted in neither N nor Q ; the seven metrics and D leave it out; no verb changes one, and none takes one as an operand except that T 's actor may be V_{derived} , read as it is, though a verb may read one, as Assembly reads V_{derived} and the seat; and it needs no τ -justified admissibility of its own ([Axiom 0](#)), the status V_{derived} already has. It is not a coordinate: the four coordinates and the sort tag are unchanged. Meta vectors ride

with the cores they describe. The paper has used the idea already, as the metadata on a walk’s steps; this subsection names it for atoms.

Form. A meta vector is a vector, and nothing constrains its form: it may need one coordinate or many. V_{derived} takes as many as the stage has, and a location takes two angles. A *scalar* meta vector, a time or an expiry, takes a designated axis of the core it labels and carries its value as the coordinate on that axis, uncapped, so that the vector distance between two of them is the difference of their values; a kind that needs more information may use a plane or more. Meta vectors share their core with the information vectors and travel with it. Turn an atom as a whole — T acts on one occupant, so the same rotation and phase shift is applied to each of its vectors — and everything it holds moves rigidly with it: V_{derived} of the turned atom is the turned V_{derived} , while a scalar meta vector, whose value is a coordinate on an axis of the atom’s own frame, is carried unchanged. Angles, magnitudes, and distances among the information vectors and V_{derived} are preserved, so turning an atom changes where it points and not what it says. Which axis suits which kind is a recommendation, the business of the catalogue of Section 6.5, and not an axiom.

Two are already in the paper. V_{derived} (§2.10) is the sum of an atom’s informational vectors, computed from them and so recomputed whenever the content is, and read only over the cores valid at the reading time (below). It exists per core and per atom, and the atom’s is exactly the sum of its cores’, since $\sum_i r_i \hat{n}_i$ regroups by core. The seat $\vec{P}$ (Section 9.1) is the other: uncapped, outside the four-tuple, and entering only the separation d_{AB} .

The recorded ones. The rest are recorded when a core is born, not computed.

- $V_{\text{timestamp}}$, per core, is when the core was born, an absolute (non-cyclic) time, so that age, now — $V_{\text{timestamp}}$, is derived and never stored: a stored age would be wrong the moment time passed. “Now” is external context, as the lexicon is. A clock-face vector, minute on ϕ , hour on χ , θ held at the equator, may sit beside it for grouping by time of day. It cannot replace it: it returns the same vector for now, a day ago, and a week ago. Putting the hour on θ fails differently: at hour 0 every minute lands on one pole, and 11:59 and 0:00, a minute apart, land antipodal.
- V_{expiry} , per core, is when the core stops being valid, an absolute time. Its absence means no expiry. A core is valid exactly when V_{expiry} is absent or now $< V_{\text{expiry}}$.
- V_{location} , per core and, where a use supplies one, per atom, is a point on a sphere, latitude as the polar angle and longitude as the azimuth, the structure $\hat{n}$ already has. It is not a sort: the cap $r \leq 1$ encodes the strength of content, and a place is a coordinate describing where content applies, not a claim with a strength. That is the line already drawn for the seat. A location *sort* would be a unit in its own right and would have to pass Axiom 0; a meta vector need not.
- V_{link} , per core, names another atom by one of that atom’s meta vectors — an identifier such as the identity key of UnCollapse (Section 3.4), or any other — and never by its content. A word is never linked to a poem: the words do not know the poem, the path knows it. Only backward links are stored, at birth: a result

names its sources, and the older atom is not edited. Forward and bidirectional links are views, the search for atoms whose backward link names a given one. Links are how sequences, graphs, and layered clusters can be recorded among atoms; building them is left to Assembly and to Inference.

Absence is not zero, for any meta vector. An absent location is not the point $(0, 0)$, and a sentinel such as “expiry 0 means none” would read as already expired under $\text{now} < V_{\text{expiry}}$.

Where they are held. Meta vectors are held per core, and each core keeps the ones it was born with. A Collapse result, as a new atom, also carries meta vectors of its own, made when it was made — a new $V_{\text{timestamp}}$, and a backward V_{link} naming its sources — beside the cores’ own; so an atom’s own birth and its cores’ births are separate facts. The same holds for every result, not Collapse alone: a result is a new atom with its own new $V_{\text{timestamp}}$ and a backward V_{link} naming its operands, and the cores that survive into it keep theirs. A result is, for these purposes, one new core on top of the cores it keeps: besides its timestamp and backward link it carries its own V_{derived} , the sum of everything it holds, and it may carry its own V_{location} and V_{expiry} where a use supplies them; absent means absent. Collapse destroys none of the cores’ meta vectors, and no verb computes a location or an expiry for a result from those of its operands.

Identity and containment. An atom is its vectors together with their grouping into cores; the grouping is part of its value, and how it is laid out in storage is not this paper’s concern; the meta vectors are the labels. Two things in an atom are easy to mistake for each other, and only one is a meta vector. A core’s *identity* is read from meta vectors it already carries: the identity key, the tuple of its sort, its source link and its timestamp. It names the core, for UnCollapse and for any use that wishes to find it, no new kind of meta vector is needed, and it never enters equality. Its *containment* is not a meta vector. The containment vector records which core holds which vector — one entry per vector, naming the position of its core in the atom’s list of cores (Axiom 7), a label that says only which vectors share a core, so that renumbering the cores changes nothing — and so is the grouping itself, part of what the atom is: equality compares it, and a meta vector never enters equality. The identity can be dropped, and UnCollapse still works by position; the grouping cannot be dropped without making a different atom.

The atom as a data unit. Besides what the verbs read from it, an atom is a unit of stored data, and it can sit beside the familiar containers. It does not replace them. Each has a counterpart here:

Container	Counterpart	What it keeps
set	a bowl, by Bundle (Axiom 4)	which atoms are present; no repeats, no order
counted bag (multiset)	a Collapsed atom (Section 3.3)	every vector, repeats counted; no order
list, tuple	a path or tape (Sections 6.3 and 4.5)	order
dictionary	the cores of an atom, keyed by their identity keys	for each key, the vectors that core holds

Three limits belong to the statement. An atom keeps no order of its own; order lives in the path. Its values are vectors, found by their identity key or by comparison and not

by an index. And reading an atom back as a name, a token, say, is a comparison with a seated unit, not an exact inverse, which a use supplies (the companion paper on Inference takes this up).

Expiry is a reading. Everything is read at a time t , by default now, and fixing t keeps every law deterministic. A core is valid at t exactly when t is earlier than every expiry it carries. An atom’s own expiry applies to every core in it, and a contribution’s expiry is carried onto each of its cores when Collapse keeps them, beside any of their own. An expired core contributes no vectors: it reads as a black core, and its meta vectors take no part either. The counts of the stored atom, N and D , do not change with t , while Q , V_{derived} , the metrics, and every process read only the valid cores. The atom itself is never changed: expiry is a reading and not an edit. An expired core is noise, and removing it is optional: UnCollapse, selected by expiry and repeated, takes each such core out with the vectors it holds and gives a new atom, in which N and D then fall. A backward link is lineage, not participation, and stays readable after its source expires.

Atoms are not edited. An atom is a value. Change is Collapse or UnCollapse, each returning a new atom and leaving the original unchanged, and each core keeps the meta vectors it was born with. A recorded meta vector therefore never changes after birth, and there is no “last modified” meta vector, because nothing is modified: a change is a new atom.

Equality. Two vectors are equal when their coordinates agree, and two cores when they hold equal vectors. Two atoms are equal when their cores can be paired off so that paired cores are equal; so grouping counts, and two atoms with the same vectors grouped differently are not equal, as UnCollapse, N and D show (Section 3.3). Meta vectors never enter this equality, the one $\oplus$, D and UnCollapse use. Two individuals are the *same individual* when they are equal and their meta vectors agree too: the stricter test, as $==$ differs from “is” in a program. Two cores of equal content born at different times are equal but not the same individual: the identity key tells them apart and the value does not. Were the timestamp part of the value, every core would be unique by birth time, $N - D$ would always be 0, and $A \oplus A = A$ would never apply: three contributions A , B , A have $D = 2$ by content and $D = 3$ with timestamps. Equality is exact. Measured readings are made comparable when they are seated, by rounding to a stated resolution or by seating a window of readings as one core (grain, Section 3.3); the techniques belong to later papers. A bowl may carry meta vectors, and $\oplus$ does not read them; anything that wishes to may.

Nothing loses identity. Under Collapse, Bundle and Mix the individual persists: what each atom holds keeps its own meta vectors. Collapse keeps every core inside its result. Bundle gives value-equal individuals one seat, discarding the count as content, and keeps their meta vectors there, where the count survives if they differ. Mix is a reading and does not consume its sources, which stay exactly as seated; its result is a new atom whose value fuses theirs and which names them by backward links. This is persistence, not recovery. UnMix remains impossible, since the sources are not determined by the result (Section 3.2), so what survives a Mix is the sources and the result’s link to them, never a way back from the result alone.

What they are for. Meta vectors group atoms for a task: atoms with alike meta vectors can be collected, ordered, or assembled together. The role is read-only: no verb changes a meta vector, and of them only V_{derived} may act, as itself (Axiom 7). Grouping needs a comparison for each kind, stated with the kind: great-circle distance for a location, cyclic distance for a clock face, equality for an identifier, and none is common to all. V_{derived} guides assembly through its magnitude and direction, which already stand in for the atom wherever a weight or a direction is required (§2.10). Direction is the cleaner rule: the energy’s bracket makes aligned pairs pull exactly twice as strongly as opposed ones at every separation, while magnitude enters as a weight whose effect depends on separation — at contact the best partner of an atom of strength 0.60 is weaker (0.30), and at a separation of 3 it is the strongest available. The direction is undefined where the vectors cancel, which is where $\mathcal{I}_{\text{coh}}$ is zero. Links declare structure and Assembly’s energy produces it from geometry and a seat; the two are separate. Links held in meta vectors can carry an order past $\uplus$, which does not read them, so order still comes from outside and the path is still where it is walked.

3 Process Principles

Each axiom above fixes a structure. This section states how that structure is produced and undone: the pipeline that chains Bundle and Transform into a single process, and the process-level counterpart of each combining verb — Mix and Collapse alike — including the two cases, UnCollapse and UnMix, where taking a part back is, respectively, defined and proven impossible.

3.1 Process principle: Bundle, path, result

The A_i below are already seated units, each loaded from an observed sequence by E_L (Equation (40)); Bundle is the act that discards the order and count of that sequence, not the act that produces the units themselves. With that understood, Pattern Arithmetic proceeds in three named acts, kept separate so that a bowl may grow before it is read:

$$\underbrace{A_1 \uplus A_2 \uplus \cdots \uplus A_n}_{R = \text{the bowl}} \longrightarrow T_{w_k} \circ \cdots \circ T_{w_1}(x) \longrightarrow \text{Result}. \quad (19)$$

Here x is the register occupant; the w_i are actors, taken from the bowl or from the tape. T does not act on the set R as a set. $\uplus$ *collects*: identical tokens do not double ($A \uplus A = A$), and opposite marks may sit together until a later reading. T is an ordered path, not a Mix. The result is whatever that path (or another named reading) makes of the bowl — a poem, a report, a staged endpoint.

Paint is Mix ($\boxplus$), a reading of peer states at a common direction, not a T :

$$\text{red} \boxplus \text{green} = \text{yellow}.$$

The same pair may take T_{poem} instead. That result is a verse about the two colors, not a hue — unless the poem itself walks back to yellow. Shells do not own a pathway. A verse is another path. One bowl, several ways to read it. One may collect and read in a single gesture; the algebra does not force the merge. This principle governs *order of acts*, not a new formula for a 4-tuple.

Levels are closed. A result may be seated again. Equation (19) returns a state, and a state may take a seat in a further bowl, so the three acts repeat at a higher level: words Bundle and are walked into a line; the line is seated; lines Bundle and are walked into a verse. Two properties hold at every level and are worth stating once. First, a unit carries no representation of what it constitutes. A seated word does not know the poem, and T leaves τ_x invariant (Section 4.3), so being acted upon does not change what a unit is. Second, a bowl carries no representation of the reading that will be applied to it; that is what *one bowl, several ways to read it* means above.

The consequence is a design rule rather than a theorem. Whatever a lower unit would have to know about a higher one belongs at the higher level and not in the 4-tuple: which units were taken as co-present, over what interval, within what region, and under what question. Those are constituting facts of a level, not coordinates of an atom, and keeping them out is why the atom stays at four coordinates and one sort tag while the levels above it may be as elaborate as a problem requires. A meta vector (Section 2.11) is no exception: it records a fact about the atom itself — when it was born, where it sits, what it came from — never what the atom belongs to, and it adds no coordinate. This paper works a single level and states the mechanism for the next; it does not carry a multi-level example.

3.2 Process principle: UnMix

Named here for completeness of the verb inventory, not newly proven: UnMix (Ξ), an inverse of Mix, does not exist: Mix sums a bowl into one state, and a sum forgets which terms made it.

Ξ : not possible.

This is a statement about the reading, not about the sources. Given only a Mixed result, the sources are not merely hard to recover; they are not determined by the result at all, verified directly in Section 7.2: two genuinely different input pairs — the paper’s own red-and-green pair, and a second pair at 10° and 100° — reach the identical phasor sum to machine precision, and a wrong assumed composition reconstructs the observed result exactly, its residual indistinguishable from the correct pair’s own residual of zero. An algebra that states only forced, verified identities has no honest place for an unconditional operation whose wrong answer looks exactly like its right one. Where the sources were kept, nothing was forgotten: in a bowl or a Collapsed atom, taking one back is Remove or UnCollapse, and a backward link names sources that persist. Where they were not, a recovery that adds constraints — a known number of sources, a stated prior — is conditional, and is a question of inference, which Section 7.2 already leaves to the companion paper, not a verb of this algebra. Ξ is recorded as a name for this already-settled fact, standing beside Collapse’s own UnCollapse so that the full inventory — Bundle, Remove, Collapse, UnCollapse, Mix, UnMix — has one entry per verb, proven or disproven, with nothing left unnamed.

3.3 Process principle: Collapse

Bundle discards order and count on the way to a bowl. A second, separate process does the opposite: several already-seated units may combine into one atom, with every contribution kept. Call this Collapse ($\oplus$). Where Axiom 7 states what such an atom *is* — N cores of informational vectors under one closure — this section states the operation

that *produces* one from separately seated units, and settles a question that structure alone does not answer: what Collapse does with repetition.

$$A_1 \oplus A_2 \oplus \cdots \oplus A_n = A_{\text{collapse}},$$

where A_{collapse} is the atom of Axiom 7 holding every core of every contribution, each core exactly as its source held it and none merged with another, every vector staying in the core that held it, so $N = \sum_i N_i$ and $Q = \sum_i Q_i$, whatever sorts the contributions carry. For teaching atoms $N_i = Q_i = 1$ and $N = Q = n$. Each core keeps the meta vectors it was born with, and any expiry its contribution carried (Section 2.11). Collapse, like Mix, is a reading: it does not consume its sources. $A_1, \dots, A_n$ remain exactly as seated as they were, in a bowl or otherwise; Collapse produces an additional atom alongside them, not in place of them — the same principle already stated for Mix’s own reading of a bowl, and the precondition that makes the backward link of Section 2.11 coherent at all: a pointer back to sources that had been consumed would point at nothing.

What Collapse keeps. Every vector, every count, and each core’s own meta vectors survive in the result, which is the sense in which Collapse is lossless. Sources that agree in content and in every meta vector are interchangeable, and nothing distinguishes them because nothing needs to; sources that differ in any are told apart by them (UnCollapse, below).

Collapse is not idempotent. $A \oplus A \neq A$ for every A , black included (below). Collapsing A with itself doubles every count — two copies of each of A ’s cores, not one; for a teaching atom, $N = 2$: two cores, both reading A ’s own $(r, \hat{n}, \chi)$. Shorthand for this case, used below and throughout: $A^{(k)}$ for k identical copies, so $A \oplus A = A^{(2)}$ and $A^{(k)}$ has kN_A cores. This is deliberate, and distinguishes Collapse from Bundle on the one point where structure alone leaves a choice open. Three considerations fix it.

First, Collapse’s own stated purpose is that no contribution is lost. Were $A \oplus A = A$, the second contribution would be discarded exactly as Bundle already discards a repeated member ($A \uplus A = A$) — leaving nothing to distinguish the two verbs in this case, when the reason for having both is that one discards multiplicity and the other does not.

Second, idempotence here is inconsistent with UnCollapse under its own, otherwise solid base case (below): if $A \oplus A = A$, then undoing one contribution must return the other, $A \text{ UnCollapse } A = A$, contradicting $A \text{ UnCollapse } A = 0$ — which holds independently, matching $\{A\} \setminus A = \emptyset$ at the atom level. Both cannot stand unless $A = 0$.

Third, and most directly: a construction elsewhere in this paper already needed fixing for exactly this failure mode. Two independent witnesses that happen to agree exactly must remain distinguishable from one witness stated once, on pain of repeating the evidence-count blindness that a clip-versus-bijection choice was built to correct (companion paper, *Inference from Bowls and Paths*). Idempotent Collapse reintroduces that same blindness in a different verb.

A silent contribution is still a contribution. Black is the empty seat of Axiom 2 — $N = 1$, $Q = 0$ — and Collapse keeps every contribution, so $A \oplus 0$ holds one more core than A and exactly the same vectors: an empty core adds to N and not to Q . Black is therefore not neutral under Collapse, and $0 \oplus 0 \neq 0$: it holds two cores and no vectors. For $A = 0$ the second consideration above is silent, since its two sides coincide;

the counting rule separates them instead, for the reason of the first. The vectors, and so every strength-based quantity, are unchanged by a silent contribution; only the count of cores moves. An expired core (Section 2.11) is the same case at its reading time: counted in N , absent from Q .

Where N and Q are kept. No new field. N , the number of cores, and Q , the number of informational vectors, are both properties of the atom's own list. N counts cores whether or not they hold vectors; Q counts only vectors. Collapse adds both counts of its operands, and an empty core adds to N and not to Q . Meta vectors are counted in neither. The counts do not say which vector sits in which core: two atoms with the same vectors and the same N and Q may group them differently, and UnCollapse of one core then leaves different atoms. The grouping, the containment vector of Axiom 7, is part of the atom's value, and equality compares it (Section 2.11). The metrics of Axiom 7 read all of an atom's vectors and ignore which core holds each.

Grain. The grain of an atom is the partition of its vectors into cores: fine when each core holds one reading, coarse when a core holds a window of them. UnCollapse acts along the grain: it takes back a core, never less. A core may hold any number of vectors, and the seating — the process that supplies the atom (Axiom 7) — chooses how many, so a ten-minute atom may be ten cores of one reading or one core of ten, and these are different atoms. Grain is fixed when an atom is seated. It can be made coarser by seating again from the vectors, which is many-to-one: the metrics and V_{derived} do not change, N and D do, and nothing records how the vectors were grouped before. No verb can make it finer; that needs the finer atom, if it is kept, or new readings. Seating again is the paper's own promotion of a result to a unit of the next level (Section 3.1).

What N does not distinguish, and what would. N counts every core, duplicate or not — Collapse's whole point. A finer question is already answerable from what is defined: of those N , how many are genuinely distinct values, as opposed to repeats of one already present? Write D for that count, the number of distinct cores among the N : two cores count as one value exactly when $\uplus$ would seat them together, which for a single-vector core means equal $(r, \hat{n}, \chi)$, and any two empty cores coincide; meta vectors never enter. D is not new machinery: it is the size of the bowl obtained by seating each core as a unit and Bundling them, since $\uplus$'s own idempotence ($A \uplus A = A$) already discards nothing but repetition. The two counts relate exactly: $D \leq N$, with equality exactly when every core is distinct, and $N - D$ is precisely how many times some core repeated. $A^{(k)}$ for a one-core A is the sharp end of this relationship: $N = k$ copies of one core, $D = 1$.

A borrowed name for the pair, offered as an analogy and not a formal claim: N reads like a nucleus's mass number, every core counted as a nucleon; D reads like its atomic number, counting only what is chemically distinct, the distinct cores. Pushed further the analogy breaks — a neutron is a genuinely different particle from a proton, not a repeated one, and nothing here claims otherwise. What transfers is the shape only: one count that keeps repetition, one that discards it, related by a fixed difference.

A worked count. Let A be an atom of one core holding two vectors, $r = 0.8$ and $r = 0.4$, and B a teaching atom with $r = 0.6$. Then $A \oplus B \oplus A$ has $N = 3$ cores and

$Q = 5$ vectors, of which $D = 2$ cores are distinct, so $N - D = 1$ repeat. UnCollapse one copy of A and $N = 2$, $Q = 3$, $D = 2$: N fell by one and Q by the two vectors that core held. Add a silent contribution instead, $A \oplus B \oplus A \oplus 0$, and $N = 4$, $D = 3$ while $Q = 5$ and no strength changes.

Collapse across differing sorts. Collapse takes atoms of any sorts. Each vector carries its own sort τ_i (Axiom 7), so two atoms differing only in sort Collapse into one atom of two cores, as they Bundle into a bowl of two seats. Nothing is combined, so no relation between the sorts is claimed (Axiom 0), and a sort inadmissible for Tier 1 may still be held at Tier 0. The atom's V_{derived} and its seven metrics read every vector whatever its sort; what one sort contributes is read from the sub-atom of its vectors (Section 3.4). Whether the whole-atom reading means anything is for the use that reads it (Section 2.11).

3.4 Process principle: UnCollapse

UnCollapse ($\ominus$) returns a Collapsed atom without one core, reducing N by exactly one and leaving the original unchanged:

$$A_{\text{collapse}} \ominus A_k = A'_{\text{collapse}},$$

where A_k names the core removed (for a teaching-atom contribution, the contribution itself) and A'_{collapse} holds the remaining $N - 1$ cores. $N \rightarrow N - 1$ is the general rule, uniformly, at every $N \geq 2$ — UnCollapse is not a reset to $N = 1$ regardless of starting count; it removes exactly the one core named, with every vector the containment vector assigns to it and nothing else, and Q falls by that core's vector count, one for a single-vector core. UnCollapse acts along the atom's grain: a core is the least it can take back.

Two cases close the ladder at its ends. At $N = 1$, removing the atom's only core does not make the atom vanish: it leaves black, the empty seat of Axiom 2 with $N = 1$ and $Q = 0$, the one exception to $N \rightarrow N - 1$:

$$A \ominus A = 0 \quad (\text{any one-core } A),$$

matching $\{A\} \setminus A = \emptyset$ at the level of a single bowl member. At $N = 2$ down to $N = 1$, an atom holding one remaining single-vector core *is* the teaching atom of that vector (Axiom 1's own $Q = 1$ base case), so

$$A^{(2)} \ominus A = A^{(1)} = A,$$

correctly, because $2 - 1 = 1$ — not a general property of UnCollapse, but the one case where its arithmetic happens to land there. No further contradiction remains once Collapse's own idempotence is fixed above.

Naming the core. Removing " A_k " presumes a specific core is named, the same way $\{A\} \setminus A$ presumes a specific member. A core is named by its *identity key*, a tuple of meta vectors it already carries — its sort, its source link and its timestamp (Section 2.11) — so no new kind of meta vector is needed, and the key never enters equality. Where every core differs in value, the value names it. Where two cores agree in value but differ

in key, the key names it. Where two cores agree in value and in every meta vector, they are the same source twice, interchangeable, and removing either gives the same atom. Where a key matches cores that differ in content — two readings seated under one label — the key cannot say which is meant, and UnCollapse takes the core’s position instead: its place in the atom’s list of cores, counting from 0, unique inside the atom and meaningless outside it. Collapse and UnCollapse renumber positions: Collapse shifts the second atom’s positions by the first’s N , and UnCollapse of position k lowers every position above k by one. A position needs no meta vector, so UnCollapse by position works on an atom whose meta vectors have been dropped. A series of cores is taken back by repeating UnCollapse, once for each. Each step gives a new atom with its own new timestamp and a backward link naming its operand, and the cores that survive keep theirs (Section 2.11).

4 Transformation Algebra: From Identity to Expression

4.1 Motivation: Two Layers of Meaning

The operators established so far ($\uplus$, $\setminus$) answer the question *who is present?* Combining Sky, Sun, and Wind under $\uplus$ yields one order-independent bowl — a bag of seats, not a blended hue. Paint is Mix ($\boxplus$), a later reading. Analogies such as Sun-Day-Moon are *not* gathers; they are deferred to the offset operator of Section 5. Bundle $\uplus$ does not, and structurally cannot, distinguish a poem in which the sun rises through a wind-blown sky from a poem in which wind extinguishes a darkening sun — both share the identical bowl $R = \text{Sky} \uplus \text{Sun} \uplus \text{Wind}$, since $\uplus$ is commutative and idempotent by Axiom 4.

The distinction between these two poems is not a difference in *ingredients*; it is a difference in *how one state acts upon another, and in what order*. This is a *relational* operation, and it requires a second operator, T , layered on top of $\uplus$: where $\uplus$ builds nouns, T builds sentences. A poem is not the bowl R itself, but a specific trajectory traced across $\mathcal{S}$ by successive transformations applied to a register occupant.

4.2 Formal Definition of the Transformation Operator

Rather than introduce T as an unrelated new primitive, it is defined using the same rotational machinery already given in §2.8: every state $w \in \mathcal{S}$ acts on another state $x \in \mathcal{S}$ by rotating x ’s direction about w ’s own axis, by an angle proportional to w ’s strength, while additively shifting x ’s phase by w ’s phase:

$$T_w(x) = \left(r_x, \text{Rot}_{\hat{n}_w}(\alpha(r_w))\hat{n}_x, (\chi_x + \chi_w) \bmod 2\pi, \tau_x \right) \quad (20)$$

where $\hat{n}_w$ is the unit axis of w , $\text{Rot}_{\hat{n}}(\alpha) \in \text{SO}(3)$ is the rotation of §2.8 about that axis by angle α , and $\alpha(r_w) = \pi r_w$ is the simplest monotonic choice mapping strength to rotation angle (a full-strength unit, $r_w = 1$, rotates by the maximal 180°). The same formula serves an actor of strength above 1, which can only be the uncapped V_{derived} of Axiom 7 read as it is; the angle πr_w is then taken modulo 360° , and for $r_w \leq 1$ nothing changes. Magnitude r_x and sort τ_x are left invariant: a transformation reorients and recolors a state, it does not strengthen or weaken it, and it does not recast it — those remain the roles of Mix (strength) and of an explicit cast (sort).

4.3 Derived Properties

The following properties follow from Equation (20) rather than being separately postulated.

Identity. If $w = \mathbf{0}$ (the Null State of Axiom 2), then $\alpha(0) = 0$ and the phase shift is 0, so $T_0 = \text{id}$.

Invertibility. The inverse of T_w is not “the same axis and the same $+\alpha$ again.” Same axis, same positive angle, repeats the turn. Inverse is $-\alpha$ on the same axis, or $+\alpha$ on the antipodal axis, together with $-\chi_w$. Define

$$w^* = (r_w, -\hat{n}_w, -\chi_w \bmod 2\pi, \tau_w), \quad (21)$$

where $-\hat{n}_w$ is the antipode ($\pi - \theta_w, \phi_w + \pi$). Then $T_{w^*}(T_w(x)) = x$ for all x . As rotations, only a multiple of 180° equals its own reverse ($r_w = 0$ and $r_w = 1$ within $\mathcal{S}$, and every whole number above); the inverse state w^* is still required so that phase is cancelled too. This undoes a transformation on a state that persists. It is distinct from $\{A\} \setminus A = \emptyset$ (last shell removed, leaving the empty bowl) and from Mix of equal anti-phase marks (a reading into black). Neither of those is an inverse of T .

Associativity. Since each T_w is a function $\mathcal{S} \rightarrow \mathcal{S}$, composition of transformations is associative automatically, by the associativity of function composition; no additional axiom is required.

Non-commutativity. Rotations about distinct axes in $\text{SO}(3)$ generically do not commute: for a, b with distinct directions (θ, ϕ) ,

$$T_a(T_b(x)) \neq T_b(T_a(x)). \quad (22)$$

This is not an additional assumption; it is inherited directly from the well-established non-commutativity of $\text{SO}(3)$, already invoked by §2.8. It is this property, and only this property, that allows two poems built from the same bowl to diverge: order of application is meaningful even though the underlying identity is not.

4.4 Beyond three dimensions

Equation (20) needs only $\hat{n}_w$ to fix a rotation because $\text{SO}(3)$ ’s generators happen to form a 3-dimensional space, exactly matching $\mathbb{R}^3$ — not because one vector generally suffices to specify a rotation. It does not: a *simple* rotation, one that fixes as much as possible and turns a single remaining plane, fixes a subspace of dimension $N_{\text{dim}} - 2$ and rotates its 2-dimensional orthogonal complement. The ladder climbs one dimension at a time. A circle turns about a fixed centre; stacking that centre through the circular slices of a sphere gives the sphere’s 1-dimensional axis; stacking again gives the 2-dimensional axis plane of the sphere in four dimensions, and again the 3-dimensional fixed space of the sphere in five:

Space	Sphere	Fixed part (the “axis”)	What turns
$\mathbb{R}^2$	S^1 , a circle	a point, dimension 0	one plane
$\mathbb{R}^3$	S^2	a line, dimension 1	one plane
$\mathbb{R}^4$	S^3	a plane, dimension 2	one plane
$\mathbb{R}^5$	S^4	a 3-D space, dimension 3	one plane
$\mathbb{R}^{N_{\text{dim}}}$	$S^{N_{\text{dim}}-1}$	dimension $N_{\text{dim}} - 2$	one plane

At every rung the fixed part grows by one dimension and the turning part stays a single plane. (A sphere is named by its own dimension, not by the space around it: the sphere in four dimensions is S^3 .) Three dimensions is not where axes stop needing more than a point — it is where they still fit in one vector.

For $\mathcal{S} = S^{N_{\text{dim}}-1} \subset \mathbb{R}^{N_{\text{dim}}}$, w supplies the fixed subspace F directly, as $N_{\text{dim}} - 2$ mutually orthogonal informational vectors from its own Collapsed atom ([Axiom 7](#)) rather than as one $\hat{n}_w$:

$$T_w(x) = \left(r_x, \hat{n}_x^F + \text{Rot}_{\alpha(r_w)}(\hat{n}_x^P), (\chi_x + \chi_w) \bmod 2\pi, \tau_x \right) \quad (23)$$

where $\hat{n}_x^F$ is $\hat{n}_x$ ’s component in F , left untouched, and $\hat{n}_x^P$ its component in the remaining 2-dimensional plane $F^\perp$, rotated within that plane by $\alpha(r_w) = \pi r_w$ exactly as before. This requires $N_{\text{dim}} - 2$ mutually orthogonal arrows in the actor’s Collapsed atom, so $Q \geq N_{\text{dim}} - 2$ is necessary but not by itself sufficient: an atom acting as w in more than three dimensions needs a Collapsed atom rich enough to supply its own axis, not just a direction. At $N_{\text{dim}} = 3$, F is one vector and Equation (23) is Equation (20) exactly — checked directly, not merely argued: the same inputs give the same output to machine precision.

Stage	Orthonormal arrows required on the <i>actor</i>
S^2 (3-D)	1 — the teaching atom, unchanged
S^3 (4-D)	2
S^{11} (12-D)	10

A teaching atom, $Q = 1$, is therefore a legal actor only at $N_{\text{dim}} = 3$; this is why three dimensions is the standard this paper keeps, not an arbitrary default — a single arrow can walk without a Collapsed atom only there.

Checked beyond the coincidence of matching 3D: invertible by $-\alpha$ on the same w at every angle tested from 0 to 359° (30°, 45°, 90°, 135°, 180°, 200°, 270° and 359°), not only small ones — the naive alternative, rotating in the plane spanned by $\hat{n}_w$ and the part of $\hat{n}_x$ orthogonal to it, positively toward $\hat{n}_w$, so that the plane and its orientation are borrowed from x instead of w , is invertible only while α stays short of the angle between w and x , and silently fails past it, since the orientation borrowed from x flips once the rotation has carried x past the axis. Sourcing F from w alone avoids this: the fixed subspace never depends on what is being acted on, the way $\text{SO}(3)$ ’s axis never did. Non-commutativity for distinct actors and angle-composition for repeated ones both continue to hold, checked directly at $N_{\text{dim}} = 4, 5, 8, 12, 20$, and 50 on a configuration that needs no seed. With e_i the coordinate axes, actor w_1 fixes $e_1, \dots, e_{N_{\text{dim}}-2}$ and so turns the plane of $e_{N_{\text{dim}}-1}, e_{N_{\text{dim}}}$; actor w_2 fixes $e_2, \dots, e_{N_{\text{dim}}-1}$ and so turns the plane of $e_1, e_{N_{\text{dim}}}$; each plane is oriented from its lower-indexed axis to its higher. The occupant is $\hat{n}_x = (1, 2, \dots, N_{\text{dim}})$ normalized. Invertibility is tested at 30°, 45°, 90°, 135°, 180°, 200°, 270°, and 359°; the order gap is $|T_{w_2}T_{w_1}\hat{n}_x - T_{w_1}T_{w_2}\hat{n}_x|$ with both actors at 90°:

N_{dim}	largest inversion error	order gap
4	1.1×10^{-16}	0.683
5	1.1×10^{-16}	0.688
8	1.1×10^{-16}	0.649
12	5.6×10^{-17}	0.584
20	1.1×10^{-16}	0.489
50	5.6×10^{-17}	0.331

Inversion is exact to rounding at every angle tested, and the two orders disagree at every dimension tested. Both results stay on the unit sphere.

One choice is left open, and is a convention rather than a gap: which orthonormal basis of $F^\perp$ orients the rotation (which direction counts as $+\alpha$) is fixed by w 's own $N_{\text{dim}} - 2$ vectors, but the specific construction is not unique — no more than the right-hand rule is the only possible convention for $\text{SO}(3)$'s own cross product. Any consistent choice, stated once, suffices; §4.7 states and exercises one.

4.5 Poems as Transformation Trajectories

A poem π built from a register occupant x and an ordered sequence of transforming states $w_1, w_2, \dots, w_n$ (typically drawn from the bowl) is the trajectory

$$\pi = (x; w_1, \dots, w_n) : \quad x \longrightarrow T_{w_1}(x) \longrightarrow T_{w_2}T_{w_1}(x) \longrightarrow \dots \longrightarrow T_{w_n} \dots T_{w_1}(x). \quad (24)$$

Two poems may share an identical bowl (the same theme, the same “ingredients”) while tracing entirely different trajectories — this is the formal sense in which many poems arise from the same units of truth. Conversely, because rotation composition is many-to-one, two distinct trajectories may terminate at the same endpoint; a poem's identity is therefore carried by its full path through $\mathcal{S}$, not by its final state alone.

Unlike $\uplus$, T does not Bundle a repeated actor: $T_w \circ T_w \neq T_w$ in general, since composing a rotation with itself composes the angle, $T_w \circ T_w = T_{w'}$ for some $w' \neq w$ rather than returning T_w . A path that visits the same actor n times is therefore n genuine steps, not one step counted n times, and its length recovers that count without any coordinate carrying it. This is the sense in which the path, unlike the bowl, does not discard multiplicity (Section 2.5).

A step of the trajectory may carry more than the bare actor: $\pi = ((x; w_1, m_1), \dots, (w_n, m_n))$ for some label m_i — a discrete time bucket, say. This changes nothing about the atom: m_i is metadata on the walk, not a coordinate of any x or w_i , and Equation (24) runs exactly as before on the w_i alone. What such a label buys, and does not buy, follows directly from the discreteness point above: a handful of buckets behaves like a sort tag and does not block $A \uplus A = A$ where it applies, but a bucket is a fact about how a particular walk was assembled, not a property its occupant carries into the next one.

4.6 Worked Numerical Example: Sun, Wind, and Sky

Let Sky be the state acted upon, with direction $(\theta, \phi) = (45^\circ, 45^\circ)$, i.e. Cartesian axis $(0.500, 0.500, 0.707)$. Let Sun define an axis along $(\theta, \phi) = (90^\circ, 0^\circ)$, i.e. $(1, 0, 0)$, with strength $r_{\text{Sun}} = 1.0$, giving rotation angle $\alpha_{\text{Sun}} = 180^\circ$. Let Wind define an axis along $(\theta, \phi) = (90^\circ, 90^\circ)$, i.e. $(0, 1, 0)$, with strength $r_{\text{Wind}} = 0.7$, giving $\alpha_{\text{Wind}} = 126^\circ$.

Order 1 — Sun acts first, then Wind (the sun rising through a wind-blown sky):

$$T_{\text{Wind}}(T_{\text{Sun}}(\text{Sky})) \approx (-0.866, -0.500, 0.011) \quad (25)$$

Order 2 — Wind acts first, then Sun (wind extinguishing a darkening sun):

$$T_{\text{Sun}}(T_{\text{Wind}}(\text{Sky})) \approx (0.278, -0.500, 0.820) \quad (26)$$

The two endpoints are clearly distinct, confirming Equation (22): identical primitives, identical bowl, opposite structural readings. Notably, if $\chi_{\text{Sun}} = 0^\circ$ and $\chi_{\text{Wind}} = 90^\circ$, the accumulated phase shift is 270° in *both* orderings, since phase addition is commutative. The two poems therefore share the same blended color/mood even as their spatial — structural — meaning diverges. This asymmetry (shared affect, divergent structure) is a numerical check of the algebra, not a theory of verse.

4.7 Worked Numerical Example: A Four-Dimensional Actor

The construction of §4.4 is checked here by hand, not only by claim. Let $N_{\text{dim}} = 4$. An actor w supplies its required $N_{\text{dim}} - 2 = 2$ orthonormal arrows as the first two standard basis vectors, $e_1 = (1, 0, 0, 0)$ and $e_2 = (0, 1, 0, 0)$, fixing $F = \text{span}(e_1, e_2)$ and leaving the active plane $F^\perp = \text{span}(e_3, e_4)$. Take $r_w = 0.5$, so $\alpha(r_w) = 90^\circ$. Let the occupant be

$$\hat{n}_x = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}}, 0 \right),$$

a unit vector with a nonzero component in both F and $F^\perp$, so the example shows both halves of Equation (23) at once. Its F -component, $(\frac{1}{2}, \frac{1}{2}, 0, 0)$, is exactly what T_w leaves untouched; its $F^\perp$ -component, $(0, 0, \frac{1}{\sqrt{2}}, 0)$, is what turns by 90° within $\text{span}(e_3, e_4)$, landing on $(0, 0, 0, \frac{1}{\sqrt{2}})$. Summing:

$$T_w(\hat{n}_x) = \left(\frac{1}{2}, \frac{1}{2}, 0, \frac{1}{\sqrt{2}} \right). \quad (27)$$

Applying T_w again with $-\alpha$ recovers the $F^\perp$ -component exactly, $(0, 0, \frac{1}{\sqrt{2}}, 0)$, since F was never touched to begin with:

$$T_{w^*}(T_w(\hat{n}_x)) = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}}, 0 \right) = \hat{n}_x, \quad (28)$$

confirming invertibility at a genuine right angle, not only for the small perturbations a reader might otherwise suspect were doing the work. The orientation convention of §4.4 is exercised here for the first time: $f_1 = e_3$, $f_2 = e_4$, in that order, is the specific choice this example makes, and a different ordering would rotate the other way.

4.8 Scope

T is deliberately left unconstrained beyond Equation (20): any sequence of any transforming states is a formally valid trajectory in three dimensions, or in N_{dim} dimensions provided each actor's own Collapsed atom is rich enough to name its F (§4.4); an underqualified actor is not a shorter path, it is not a legal w at all. Pattern Arithmetic in this form supplies the substrate on which expression is built — the combinatorial space of possible trajectories — and does not yet supply a grammar constraining which trajectories are coherent, well-formed, or meaningful to a human reader. Whether such constraints belong inside this framework (as further axioms on admissible sequences of T) or outside it (as a learned or externally specified grammar layered on top) is left open.

5 Analogical Arithmetic: The Offset Operator

Offset is another verb. It is not $\uplus$ (no Bundle), not $\oplus$ (no Collapse, nothing is combined into one), not Mix (no phasor blend), and not T (no spin of an actor on a path). It copies a *walk* from one seat to another. The landing is a new direction on the same globe. A dictionary may later pin a name on that landing. The walk does not mint the name.

5.1 Recipe: logarithm, carry, exponential

Write directions as unit vectors $\hat{n} \in S^2 \subset \mathbb{R}^3$. The walk from p to q is the tangent vector

$$\theta = \arccos(\hat{p} \cdot \hat{q}), \quad \log_{\hat{p}}(\hat{q}) = \frac{\theta}{\sin \theta} (\hat{q} - (\hat{p} \cdot \hat{q}) \hat{p}) \quad (\theta \neq 0), \quad (29)$$

with $\log_{\hat{p}}(\hat{p}) = \mathbf{0}$. Its length is the geodesic angle. The reverse map, from a tangent vector v at p , is

$$\exp_{\hat{p}}(v) = \cos \|v\| \hat{p} + \sin \|v\| \frac{v}{\|v\|} \quad (v \neq \mathbf{0}), \quad (30)$$

and $\exp_{\hat{p}}(\mathbf{0}) = \hat{p}$. Given three seats m, k, w (source, source-image, new start), set

$$v = \log_{\hat{n}_m}(\hat{n}_k), \quad \hat{n}_* = \exp_{\hat{n}_w}(\text{transport}_{m \rightarrow w}(v)). \quad (31)$$

The carry rotates the tangent vector by the geodesic angle about the axis of $m \rightarrow w$:

$$\theta_{mw} = \arccos(\hat{n}_m \cdot \hat{n}_w), \quad \hat{u} = \frac{\hat{n}_m \times \hat{n}_w}{\|\hat{n}_m \times \hat{n}_w\|} \quad (\hat{n}_m \neq \pm \hat{n}_w), \quad (32)$$

$$\text{transport}_{m \rightarrow w}(v) = \text{Rot}_{\hat{u}}(\theta_{mw}) v = v \cos \theta_{mw} + (\hat{u} \times v) \sin \theta_{mw} + \hat{u} (\hat{u} \cdot v) (1 - \cos \theta_{mw}). \quad (33)$$

The axis is undefined when $\hat{n}_m = \pm \hat{n}_w$. Coincident seats ($\hat{n}_m = \hat{n}_w$): the carry is v itself. Antipodal seats ($\hat{n}_m = -\hat{n}_w$): no unique short geodesic; that case is a separate convention and is not used below. For a first recipe, if m and w are not far, transport may be omitted and v used at w as written. That case is crude and marked as such. The full recipe is (33), so the heading stays a ground heading.

Strength and colour are not invented by the walk. This paper takes

$$r_* = r_w, \quad \chi_* = (\chi_w + \chi_k - \chi_m) \bmod 2\pi, \quad \tau_* = \tau_w. \quad (34)$$

The target keeps its r and its sort. The mood increment of the pair $m \rightarrow k$ is reused. Nothing here uses $\alpha = \pi r$ from T . With (31) and (34) together, offset returns a complete state, not a bare direction:

$$\text{offset} : \mathcal{S} \times \mathcal{S} \times \mathcal{S} \longrightarrow \mathcal{S}, \quad (m, k, w) \mapsto (r_w, \hat{n}_*, \chi_*, \tau_w).$$

It is therefore a verb on the same footing as $\uplus$, $\oplus$, $\boxplus$ and T : a landing may be Bundled into a bowl, Collapsed with another into one atom, walked from, or used as the start seat of a further offset. The question of *whether anything is seated there* is separate, and is settled by a dictionary, not by the walk. $\tau_* = \tau_w$ because the question asked is what stands to w as k stands to m ; the answer is of w 's kind. A walk copied between sorts would need an explicit cast (Section 6), which offset does not perform. Two seated units keep two strengths. If Day has $r = 0.9$ and Night has $r = 0.8$, both sit; there is no $0.9 + 0.8 = 1.7$. That sum is ambient $\mathbb{R}^3$ arithmetic and leaves the ball. $\uplus$ gathers the two seats. A later path may turn a dial or clip a Mix. It does not get to smash two r 's into one illegal length as the price of seating them.

Beyond three dimensions. Equation (33) is written with a cross product, which exists only in three dimensions, but what it performs is the simple rotation of §4.4: it turns the plane spanned by $\hat{n}_m$ and $\hat{n}_w$ through θ_{mw} , carrying m onto w , and fixes every direction orthogonal to that plane. Written that way it holds on any $S^{N_{\text{dim}}-1}$:

$$\begin{aligned} \text{transport}_{m \rightarrow w}(v) = v + (\cos \theta_{mw} - 1)[(v \cdot \hat{n}_m)\hat{n}_m + (v \cdot \hat{b})\hat{b}] \\ + \sin \theta_{mw}[(v \cdot \hat{n}_m)\hat{b} - (v \cdot \hat{b})\hat{n}_m], \end{aligned} \quad (35)$$

where $\hat{b} = (\hat{n}_w - (\hat{n}_m \cdot \hat{n}_w)\hat{n}_m) / \|\hat{n}_w - (\hat{n}_m \cdot \hat{n}_w)\hat{n}_m\|$ is the unit direction of w as seen from m within their plane. Unlike T , offset needs no Collapse for this: it is handed two seats, and the two seats name the turning plane themselves. At $N_{\text{dim}} = 3$, Equation (35) is Equation (33) exactly, and the equator example below lands on the same point. In four dimensions, with $\hat{n}_m = (1, 0, 0, 0)$, $\hat{n}_k = (\frac{\sqrt{3}}{2}, 0, 0, \frac{1}{2})$, and $\hat{n}_w = (0, 1, 0, 0)$, the step $v = \log_{\hat{n}_m}(\hat{n}_k) = (0, 0, 0, \frac{\pi}{6})$ lies outside the turning plane, is carried unchanged, and lands at $\hat{n}_* = (0, \frac{\sqrt{3}}{2}, 0, \frac{1}{2})$.

5.2 A check on one equator, not a definition

Place three arrows in the xy -plane:

$$\hat{n}_m = (1, 0, 0), \quad \hat{n}_k = \left(\frac{\sqrt{3}}{2}, \frac{1}{2}, 0\right), \quad \hat{n}_w = (0, 1, 0)$$

(0° , 30° , 90°). Then $\log_{\hat{n}_m}(\hat{n}_k) = (0, \pi/6, 0)$. Here $\theta_{mw} = 90^\circ$ and $\hat{u} = (0, 0, 1)$, so (33) is a quarter-turn about $\hat{z}$ and gives $(-\pi/6, 0, 0)$. The exponential at $\hat{n}_w$ lands at

$$\hat{n}_* = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right) \quad (120^\circ).$$

On this slice the walk *looks like* $90^\circ + 30^\circ = 120^\circ$. That sum is a protractor check. It is not the operator, and it is not a night. If a lexicon writes the names Sun, Day, Moon on the first three seats, it may write Night on the landing: whatever Sun is to Day, Moon is to that fourth place. The geometry only promised a fourth pin. The same moral as weather: Day and Night are dictionary pins, not sums of luminaries. The step that carries Sun to Day is a walk, and the walk is what is copied; $\uplus$ of those seats does not land on a time of day. Whether the two halves really are the same walk depends entirely on how a lexicon seated them, which is a dictionary claim and not one this paper makes. Offset only copies a walk; a later comparison may find the landing *near* an already seated name. It does not mint the name.

5.3 Offset is not T

$T_w(x)$ spins an occupant about an axis. Offset plants a new seat from three seats. A path does not need (31). A dictionary act does not need (20). Do not encode the walk as a ghost actor whose sole job is to trigger T .

The popular ambient sum $\hat{n}_k - \hat{n}_m + \hat{n}_w$ then normalize (Recipe 1) is the embedding fluke written as geometry. It is recorded as a cheaper cousin, not as a derivation of the landing. This paper uses Recipe 2 so the copied step stays on the skin of the ball.

6 Encodings, Sorts, and Mixed Documents

The unit sphere is a single bounded *register*. Numbers, words, and colors are not three different globes; they are three encodings of a token onto the same globe. A scientific paragraph already mixes those alphabets on one page. Pattern Arithmetic may do the same, provided each loaded state carries a *sort* and the operators respect that sort.

6.1 Two primary sorts

A state $x \in \mathcal{S}$ is the info atom (unit) of [Axiom 1](#). What the geometric 4-tuple (r, θ, ϕ, χ) means depends on the sort τ and the encoding used to write it:

- **Numeric sort (num).** The state represents a scalar. One toy encoding — the only numeric encoding used in this paper — stores an integer n on a marked clock as $\chi = 2\pi n/M$ (arithmetic modulo M), at a fixed direction, with $r = 1$. Another, stereo-style, is recorded as an option: it would send $x \rightarrow \pm\infty$ toward the poles so that unbounded reals still occupy the globe. It is not used below.
- **Lexical sort (word).** The state is a labeled point in a toy dictionary living on the same globe. Similarity is the angle between points (cosine similarity). A bag of words is formed by $\uplus$. A sentence or poem is formed by T .

Schoolbook addition lives only in **num** mode. Mix as a *domain* reading lives in the Tier-1 sorts of [Axiom 0](#): **color** under this paper's phasor inclusion, and **note-pitch**, whose octave equivalence makes the circle native rather than borrowed. The two sorts worked below are **num** and **word**; neither takes a domain Mix. Two **word** states may be mixed as geometry; that is not a χ -reading of language. The two equations may share a sphere. They may not share an operator.

6.2 Numeric mode: $7 + 5 = 12$ on the clock, and why $\uplus$ is not 12

Fix the modulus $M = 24$ and the encoding

$$E_{\text{num}}(k) = (r = 1, \hat{n} = (1, 0, 0), \chi = 2\pi k/24, \tau = \text{num}), \quad k \in \mathbb{Z}/24\mathbb{Z}. \quad (36)$$

Decode is the inverse chart: $k = M\chi/(2\pi)$, an integer hour mark. Then 7 sits at $\chi = 105^\circ$, 5 at $\chi = 75^\circ$, and 12 at $\chi = 180^\circ$. Schoolbook $+$ is decoded arithmetic:

$$7 + 5 = E_{\text{num}}^{-1}(E_{\text{num}}(7)) + E_{\text{num}}^{-1}(E_{\text{num}}(5)) = 12, \quad E_{\text{num}}(12) = (1, (1, 0, 0), 180^\circ, \text{num}). \quad (37)$$

That is $7 + 5 = 12$ as a calculation on $\mathcal{S}$: decode, add in $\mathbb{Z}$, encode. Bundle is legal on the same pair but answers a different question: $7 \uplus 5 = \{7, 5\}$ records which numerals are present and is not 12. If one instead applies Mix (Equation (8)) to the same two geometric marks, the phasors are

$$z = e^{i7\pi/12} + e^{i5\pi/12} = 2\cos(\pi/12) e^{i\pi/2}, \quad (38)$$

so $r' = \min(1, 2\cos 15^\circ) = 1$ and $\chi' = 90^\circ$. The clock would read that mark as hour 6, not hour 12. A bowl $\{E_{\text{num}}(7), E_{\text{num}}(5)\}$ is two seats, still not 12. This single contrast is why the type table exists: $+$ answers *how many*; $\uplus$ and Mix do not.

6.3 A document is a sequence, not a blend

Words and numbers coexist in reports because a document is an ordered tape of tokens, not one mixed arrow. Write a paragraph as a sequence of info atoms

$$(x_1, \dots, x_m), \quad x_i = (r_i, \hat{n}_i, \chi_i, \tau_i), \quad \tau_i \in \{\mathbf{num}, \mathbf{word}\}, \quad (39)$$

The sphere holds one current occupant of the register; the paragraph is the history of loads and actions. A poem (Section 4.5) is one species of that history: a T -walk on one occupant. Juxtaposition is another. Mix is not.

They may meet in only three honest ways.

Juxtaposition. The default. Tokens sit in order and no algebra is applied between them. In “the sample reached 37.4°C at dawn,” the number 37.4 is not blended with the word dawn.

Typed action. The operator $T_w(x)$ is allowed across sorts only with an explicit output sort. Two useful conventions, which the rest of this paper adopts, are:

- a word acting on a number yields a number-in-context (still **num**);
- a number acting on a word yields a refined word (still **word**).

A number may also set a numeric dial of T — for example the rotation angle $\alpha(r)$ — because that dial is already a scalar. That is how “wind of 12 m/s” can strengthen a turn without becoming a synonym of wind.

Cast. Rare and named. The map $E_{\mathbf{num} \rightarrow \mathbf{word}}(2) = \mathbf{two}$ turns a digit into a spoken numeral; the inverse is parsing. Casts are dictionary maps, not $\uplus$. A lexicon is a cast of the same kind: it sends a surface token to a state,

$$E_L(\mathbf{token}) = x \in \mathcal{S}, \quad (40)$$

one map for each language, and two lexicons may send different strings to the same state. The shared register that this makes possible is a design, taken up in the companion paper.

The illegal meeting is an untyped blend

$$\mathbf{dawn} \uplus 37.4, \quad (41)$$

which has no meaning until one side is first cast into the other’s sort.

6.4 Operator legality

Table 2 records which buttons accept which pairs. Here $+$ stands for schoolbook arithmetic on the numeric encoding (decode, operate in $\mathbb{Z}$, encode; $+$ is the worked case of Section 6.2), $\uplus$ is idempotent Bundle, $\oplus$ is count-preserving Collapse (Section 3.3; not idempotent, unlike $\uplus$), $\boxplus$ is Mix, T is typed action, and $x - a + b$ is the analogical offset of Section 5.

6.5 The meta-vector catalogue

The concept dictionaries and per-language lexica that a shared register would need are not part of this paper; the companion paper *Pattern Arithmetic: Inference from Bowls and Paths* carries them as a deferred programme. One catalogue belongs here, because the meta vectors of Section 2.11 are defined here.

All meta vectors are uncapped. A scalar kind takes one axis, $\mathbb{R}^1$; a kind of two angles takes a plane, $\mathbb{R}^2$; each lives in its own space or in axes disjoint from the content's. The entries are recommendations of the same standing as the dictionaries, so that meta vectors from different sources line up; a use may choose another unit if it states it.

Kind	Space	Unit	Compared by
$V_{\text{timestamp}},$ V_{expiry} clock face	$\mathbb{R}^1$, one axis each	seconds since 1970-01-01 00:00 UTC	difference of values
V_{location}	$\mathbb{R}^2$: minute on ϕ , hour on χ , θ at the equator	angle	cyclic distance
V_{derived}	$\mathbb{R}^{N_{\text{dim}}}$	strength	vector distance
seat $\vec{P}$	$\mathbb{R}^3$	a length the use chooses	Euclidean distance
V_{link}	$\mathbb{R}^1$, an identifier	none	equality
V_{identity}	the tuple (sort, link, timestamp) of meta vectors the core already carries; no axis of its own	none	equality; never enters the equality of atoms
$V_{\text{container}}$	$\mathbb{R}^1$ per vector: the position of its core, a label and not a measurement	an index	equality, up to renumbering the cores; the grouping itself, so it enters the equality of atoms and is not a meta vector (Section 2.11)

7 Operational Algebra & Numerical Demonstration

7.1 Mix of two shells

Collecting $r = 0.6$ at 0° with $r = 0.8$ at 120° (same place) yields two shells, not an angle. Applying Mix ($\boxplus$) and Equation (8) yields the polar chart $(r', \theta', \phi', \chi') \approx (0.721, \theta, \phi, 73.9^\circ)$. The older figure 66.6° is withdrawn: Axiom 5 as a length-only formula never produced it.

7.2 Rotational transposition, and what division is not

Rotation is the rule of §2.8: $\mathbf{R} \in \text{SO}(3)$ turns $\hat{n}$. Inverse rotation is well-defined. Division as “deconvolution and signal unmixing” — UnMix ($\boxminus$) — is not defined in this paper — shown here to be ill-posed for this algebra specifically, not merely unbuilt.

Given only a Mixed result, the bowl that produced it is not merely hard to recover; it is not determined by the result at all. The introduction’s own red-and-green example, 0° and 120° at equal strength reaching yellow at 60° , is one point in an infinite family: two components at 10° and 100° , correctly weighted, reach the identical phasor sum, to machine precision — verified directly, not merely argued, and neither pair is preferred by the result over the other. Assuming a specific composition and solving for it does not rescue this. A wrong assumed composition still reconstructs the observed result exactly: solving for 10° -and- 100° strengths against the true red-and-green result returns a residual of 10^{-17} , indistinguishable by any check internal to the algebra from the correct pair’s own residual of zero. An algebra that states only forced, verified identities has no honest place for an operation whose wrong answer looks exactly like its right one.

This is deliberately left to inference (companion paper, *Pattern Arithmetic: Inference from Bowls and Paths*), where a comparison against a witnessed reference is held to a stated confidence and may abstain — a discipline this algebra’s own Mix does not have available to it, and should not manufacture.

8 Higher-order systems as bowls and paths

These examples illustrate the $\boxplus$ -versus- T split with borrowed vocabulary. They are costumes (Section 1.4), not models of weather, ecology, or disease. Higher-order truths are bowls of basic units read by a path. A storm, a forest, or a cancellation is such a reading. The path defines the higher-order system: the bowl only lists who is present.

A vertical pinball table makes the same split visible. Each bumper is one seated unit — a direction and a size r . The set of bumpers on the board is the bowl (wind, heat, moisture, and the eye). The pockets at the bottom are dictionary pins already named storm and calm. Dropping a ball from a slot is a path T_{weather} : the ball meets some bumpers, in some order, and each kick depends on that bumper’s angle and strength. The landing pocket is a comparison after the walk. Adding the bumpers together does not *make* a pocket. A different slot, or a different order of kicks, is a different weather. This is one rough picture of *a* transform path, not the mechanics of every T , and not a second axiom. A Mix reading ($\boxplus$), and a poem path T_{poem} , need not bounce.

8.1 Atmospheric storm (costume): one pipeline with numbers

These walks are ordered readings of many co-present units that can land near a higher-order name already seated in a dictionary. Four incoming states, one bowl, two paths, two endpoints. Every figure below is Rodrigues’ rotation of $\hat{n}$ by $\alpha = \pi r$ about the actor’s axis, with χ added. A reader can recompute.

1. Input states.

$$\begin{aligned} \text{Wind} : r &= 1, \hat{n} = (1, 0, 0), \chi = 0^\circ, \tau = \text{word}, \alpha = 180^\circ, \\ \text{Heat} : r &= \frac{1}{2}, \hat{n} = (0, 1, 0), \chi = 90^\circ, \tau = \text{word}, \alpha = 90^\circ, \\ \text{Moisture} : r &= \frac{2}{3}, \hat{n} = (0, 0, 1), \chi = 45^\circ, \tau = \text{word}, \alpha = 120^\circ, \\ \text{Eye} : r &= 1, \hat{n} = \left(\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}\right), \chi = 0^\circ, \tau = \text{word}. \end{aligned}$$

The eye sits at $(\theta, \phi) = (45^\circ, 0^\circ)$.

2. Bundle into the bowl.

$$R = \text{Wind} \uplus \text{Heat} \uplus \text{Moisture} \uplus \text{Eye} = \{\text{Wind}, \text{Heat}, \text{Moisture}, \text{Eye}\}. \quad (42)$$

Four distinct names, four seats. A second Wind would not open a fifth. Call this bowl $s(\text{weather})$ if a name helps: it is still four pins, not a fifth vector. Mix is a different reading and is not the storm. In the dictionary, storm and calm are *other* seated units — separate doorbells, not sums of wind, heat, moisture, and the eye. Mere $\uplus$ cannot reach those pins. T_{weather} is the path family that reads $s(\text{weather})$; T_{deepen} and T_{clear} below are two such walks. An endpoint may fall *near* the storm pin or the calm pin. Nearness is comparison after the walk, not addition of the parts.

3. Two trajectory paths. The register occupant is the eye. The other three shells act, in order:

$$T_{\text{deepen}} = T_{\text{Wind}} \circ T_{\text{Heat}} \circ T_{\text{Moisture}}, \quad T_{\text{clear}} = T_{\text{Moisture}} \circ T_{\text{Heat}} \circ T_{\text{Wind}}. \quad (43)$$

Deepen — moisture, then heat, then wind:

$$\begin{aligned} T_{\text{Moisture}}(\text{Eye}) &= (1, (-\frac{\sqrt{2}}{4}, \frac{\sqrt{6}}{4}, \frac{\sqrt{2}}{2}), 45^\circ), \\ T_{\text{Heat}}(\cdots) &= (1, (\frac{\sqrt{2}}{2}, \frac{\sqrt{6}}{4}, \frac{\sqrt{2}}{4}), 135^\circ), \\ T_{\text{Wind}}(\cdots) &= (1, (\frac{\sqrt{2}}{2}, -\frac{\sqrt{6}}{4}, -\frac{\sqrt{2}}{4}), 135^\circ). \end{aligned}$$

Clear — wind, then heat, then moisture:

$$\begin{aligned} T_{\text{Wind}}(\text{Eye}) &= (1, (\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}), 0^\circ), \\ T_{\text{Heat}}(\cdots) &= (1, (-\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}), 90^\circ), \\ T_{\text{Moisture}}(\cdots) &= (1, (\frac{\sqrt{2}}{4}, -\frac{\sqrt{6}}{4}, -\frac{\sqrt{2}}{2}), 135^\circ). \end{aligned}$$

4. Endpoints. T leaves the occupant's strength at $r = 1$. Phase addition commutes, so both paths end at $\chi = 135^\circ$. The directions do not agree:

$$\begin{aligned} T_{\text{deepen}}(\text{Eye}) : \quad \hat{n} &= (\frac{\sqrt{2}}{2}, -\frac{\sqrt{6}}{4}, -\frac{\sqrt{2}}{4}) \approx (0.7071, -0.6124, -0.3536), \\ (\theta, \phi) &\approx (110.7^\circ, 319.1^\circ); \\ T_{\text{clear}}(\text{Eye}) : \quad \hat{n} &= (\frac{\sqrt{2}}{4}, -\frac{\sqrt{6}}{4}, -\frac{\sqrt{2}}{2}) \approx (0.3536, -0.6124, -0.7071), \\ (\theta, \phi) &= (135^\circ, 300^\circ). \end{aligned}$$

Same bowl, two weathers. The storm is the walk. It is not a fifth pin, and it is not a claim about vorticity or shear.

Further costumes (table and chair, forest, pathogen cancellation) are carried by the companion paper on Inference.

9 The Enclosure: Seats, Rest, and Camps

Every operator before [Axiom 6](#) works entirely on the bounded stage: units go in, units or states come out, and nothing asks where anything sits. Assembly asks. This section

supplies the room it asks about, and then follows what happens when units are left alone in it.

The enclosure is a costume borrowed briefly and then dropped. Many small things, left alone, falling into a few quiet piles is a familiar picture from astronomy; the units here are not stars, carry no mass, and move only by the rules below. After this paragraph the working words are *seat*, *rest*, and *camp*.

9.1 Seats

A unit, or a bowl, may carry a *seat*: a position vector $\vec{P}$ measured from one shared origin, in ordinary unbounded Euclidean space. The seat is external to the 4-tuple of [Axiom 1](#), in the same way step metadata sits on the path rather than inside the units it walks. The unit is unchanged by having one; r is still bounded, $\hat{n}$ is still a semantic address and not a physical one. For two seated units,

$$d_{AB} = |\vec{P}_B - \vec{P}_A|, \quad \hat{r}_{AB} = \frac{\vec{P}_B - \vec{P}_A}{d_{AB}}, \quad (44)$$

which completes Equation (13). Where seats exist, $\hat{r}_{AB}$ is the separation axis; no axis derived from orientation is required or used.

9.2 Rest

Seats may move; meaning does not. $\hat{n}$ and χ do not crawl, so the lexicon stays put and no unit drifts into being a different unit. What moves is $\vec{P}$, and it moves by descending the energy of Equation (13):

$$\vec{P}_i \leftarrow \vec{P}_i + \delta F_i, \quad F_i = -\nabla_{\vec{P}_i} \sum_{j \neq i} U_{ij}. \quad (45)$$

Written out, with $K_{ij} = \frac{1}{2}r_i r_j (3 + \hat{n}_i \cdot \hat{n}_j)$ and $\varepsilon_{ij} = r_i + r_j$, the force on unit i is

$$F_i = -3 \sum_{j \neq i} K_{ij} (\vec{P}_i - \vec{P}_j) (d_{ij}^2 + \varepsilon_{ij}^2)^{-5/2},$$

each term pointing from i toward j . There is no mass term and no momentum carried between steps. This is deliberate rather than an omission. The flow carries no inertia, so in continuous time an isolated pair cannot overshoot: it closes distance monotonically and never exceeds its own starting separation. The update above is explicit Euler, and it closes the distance monotonically only while δ stays below a critical step, $1/c$ with $c = 3K_{AB}/\varepsilon_{AB}^5$ and $K_{AB} = \frac{1}{2}r_A r_B (3 + \hat{n}_A \cdot \hat{n}_B)$, which for an aligned pair of equal strength r is $\frac{16}{3}r^3$. The critical step falls with strength, so no single δ suits every strength; the printed replay keeps well inside it (Section 9.4). Weighting each step by a volume-like quantity was tested and produces oscillation rather than rest, precisely where the weighting is most extreme — a ninefold difference in strength gives a 729-fold weighting under a cubed reading, and a demonstrably non-monotonic trajectory against the clean decline of Equation (45) (the printed pair of §9.4).

A direct consequence: since $F_A = -F_B$ exactly and nothing divides the resulting step, two units of any polarity move by identical magnitude toward each other. Polarity labels a bond; it does not sort the movers. A second consequence: since every force comes in an equal and opposite pair, the plain, unweighted mean of the seats is unchanged by every step.

Rest is a freeze, not a stop. Taken literally, rest arrives only once, and only at the end. The pull between two units of nonzero strength is proportional to $d(d^2 + \varepsilon^2)^{-5/2}$, which is positive at every separation $d > 0$: no two such units are ever exactly still while apart. In the continuous flow, for a finite population of fixed, nonzero strengths, given unlimited time every group joins every other and the seats converge to a single point, the plain mean of their starting seats: a seat on the edge of the cloud is always pulled inward, so the cloud's extent can only shrink, and the only configuration with no net force is a single point. The approach is asymptotic, not in finite time, and for the Euler step it is observed on the printed runs, not proved. What the enclosure offers instead is a stated freeze. A configuration is *at rest at resolution η* when

$$\max_i |F_i| < \eta, \quad (46)$$

and a configuration that has reached this threshold may be frozen and taken as the considered one — the accepted solution at that resolution. Its shape, the walk that produced it, and its camps are then what is read. The threshold η is a measurement convention, named alongside the step δ and the census cut below, and not a truth of the algebra: a finer η runs longer and may see groups merge that a coarser one froze apart. It is stated on force rather than on the size of a step for a reason. A step is δF_i , so a threshold on steps would let the choice of δ decide which configurations count as rest; a threshold on force does not.

The step δ in Equation (45) is a numerical convention and not a truth of the algebra. It sets how finely the descent is sampled, not where the freeze falls. For one isolated group this is exact: the plain mean of its seats is conserved, so it comes to rest on the plain mean of its starting seats whatever δ is used, agreeing across $\delta = 0.02, 0.01, 0.005$, and 0.002 to within 10^{-15} . Where several groups pull on one another weakly, the same four steps reach the freeze at the same time and with the same camps, and their bodies agree to within 2×10^{-7} — the ordinary discretization error of sampling the slow drift between groups (§9.4). A step chosen too coarsely relative to ε_{AB} will misbehave near close approach, in the ordinary way of any fixed-step scheme; that is a limit on the arithmetic, not on the rule.

Inside a group, the merge does not stop short of a point. Two members close together pull in proportion to their separation, so a frozen group's members sit on one seat to within a distance that shrinks with η — under 2×10^{-5} at $\eta = 10^{-5}$ in the configuration of §9.4. A frozen camp has a location and a membership. It has no internal shape.

9.3 Camps and bodies

Units left alone under Equation (45) do not always fall into one pile. The energy dies as d^{-3} , so distant units are nearly irrelevant to one another and local groups settle before they feel the far ones. Several separated groups can therefore reach the freeze of Equation (46) together, each one a *camp* — a camp at resolution η , since without a freeze they would go on to merge. The number of camps is not chosen in advance; it is whatever the rule, the starting seats, and η produce.

For each camp G_k that has frozen, its *body* is

$$\vec{P}_{c,k} = \frac{\sum_{i \in G_k} r_i \vec{P}_i}{\sum_{i \in G_k} r_i}, \quad (47)$$

the strength-weighted centre of its members. Nothing computes this during motion and no unit seeks it; each unit feels only the sum of pairwise pulls from its neighbours. The body is measured afterwards, by an observer, and is a description rather than a destination.

Where units carry inner structure, the weights r_i in Equation (47) may be replaced by $|V_i|$, the magnitude of a unit's own resultant. A unit whose inner arrows cancel still sits in the camp but barely moves its body — the body then reports surviving configuration rather than bare presence. Both readings are legitimate; they answer different questions and should not be conflated. The difference is a property of a camp read *before* its freeze. Once frozen, a camp's members sit on one seat to the resolution of η , so every weighting returns the same body to that resolution, and the body of an isolated camp is simply the plain mean of its starting seats.

A system of several camps is described by the *set* of bodies $\{\vec{P}_{c,1}, \vec{P}_{c,2}, \dots\}$, never by one grand average over all of them. Two camps ten units apart have no member at their midpoint, and reporting one body there would assert a camp that does not exist. This is the same honesty the bowl already practises when Mix returns black: where the evidence is two truths, do not mint a third that nothing occupies.

Counting camps requires a stated separation cut, or a chain of short links between members. That cut is a measurement convention and not a truth of the algebra: the camps are produced by the rule, the census of them is not.

A camp is a checkpoint, not a destination. Every camp is read at some freeze η , after some elapsed time; a longer wait, a finer η , or a different δ freezes a different, though usually nearby, configuration. Call each such freeze a *checkpoint*. Two checkpoints of the same starting seats — upstream or downstream in time, at a different η , or under a different δ — need not agree exactly, and §9.4 already prints the drifts: 2×10^{-7} between step sizes, 2.4×10^{-8} between weightings, 3×10^{-5} across a weak inter-group pull. None of these is a mistake; each is what the same rule gives at a different checkpoint.

Treating two checkpoints as *the same* freeze therefore needs one further named convention, an *allowance* ϵ_{allow} : checkpoints whose bodies differ by less than ϵ_{allow} are read as one, and checkpoints that differ by more are read as genuinely apart. This is not a new rule. It sits beside η , δ , and the census cut as a fourth measurement convention, stated once and applied consistently, not a truth of Equation (13). Every drift printed above falls well inside a stated allowance of, say, 10^{-4} ; the storm's Deepen and Clear endpoints of §8.1, 0.500 apart in direction (28.9°), do not fall inside any allowance a reader would state honestly. An allowance too generous erases real structure; one too strict reports a new camp every time the integrator takes a different step. Naming it is the price of being able to say two checkpoints are the same thing at all.

9.4 A replayable configuration

The claims of this section are printed against one configuration, so that any reader can replay them. How a configuration is generated — by hand, or by any library under any seed — is not part of the claim; the printed seats are. Six units in two groups forty apart:

Unit	r	$\hat{n}$	starting $\vec{P}$
1	0.5	(1, 0, 0)	(0, 0, 0)
2	0.6	(0, 1, 0)	(1, 0, 0)
3	0.7	(0, 0, 1)	(0, 1, 0)
4	0.8	(1, 0, 0)	(40, 0, 0)
5	0.4	$(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0)$	(41, 0, 0)
6	0.9	(0, 0, 1)	(40, 1, 0)

Under Equation (45) with the freeze $\eta = 10^{-5}$ of Equation (46), the runs at $\delta = 0.02$, 0.01, 0.005, and 0.002 take 628, 1261, 2526, and 6320 steps — the same elapsed time, 12.6, in each — and every one freezes into the same two camps, units $\{1, 2, 3\}$ and $\{4, 5, 6\}$. The bodies land at (0.33336, 0.33333, 0) and (40.33330, 0.33333, 0): the plain means of each group’s starting seats, $(\frac{1}{3}, \frac{1}{3}, 0)$ and $(\frac{121}{3}, \frac{1}{3}, 0)$, each drawn about 3×10^{-5} toward the other by the weak pull across the gap. The four runs agree on both bodies to within 2×10^{-7} ; the plain mean of all six seats is unchanged throughout; and the r -weighted and plain bodies of the first camp differ by 2.4×10^{-8} . Run alone, the first group comes to rest on $(\frac{1}{3}, \frac{1}{3}, 0)$ at all four steps, agreeing to within 10^{-15} . Neither camp is still in the literal sense: the two remain forty apart only because their mutual pull fell below η before they could close the distance.

The volume-weighting comparison of §9.2 is replayed on a printed pair: $r = 0.9$ at (0, 0, 0) and $r = 0.1$ at (3, 0, 0), both with $\hat{n} = (1, 0, 0)$, $\delta = 0.02$, 3000 steps. Under Equation (45) the separation never increases, falling from 3 to 1.80. Dividing each step by r^3 — a 729-fold weighting between the two — makes the separation increase on more than a quarter of the 3000 steps: between 830 and 855 of them, depending only on the order in which the arithmetic is carried out, because the oscillation amplifies rounding. The count is given as a range for that reason.

9.5 What the enclosure does not supply

Assembly never repels, so nothing in the enclosure pushes members apart; separated camps persist because pull dies with distance, not because anything resists, and only until a freeze is declared. There is no final plurality: left without a freeze, every camp eventually joins every other. There is correspondingly no dense center, no halo, and no sorting of members by strength — structures familiar from the borrowed picture that this rule does not reproduce, and should not be graded against. A camp does not acquire a name by coming to rest. Naming is comparison against a seated pin, which is a different question asked afterwards and elsewhere.

10 Discussion & Related Work

Each paragraph below states what is reused and what is added. None of these citations is a claim of identity. The list is a boundary-drawing exercise against formalisms that share a sphere, an angle, or a rotation (Section 1.4). Empty headings are omitted.

Bloch sphere. A pure qubit is a point on S^2 ; a mixed qubit lives in the ball. Pattern Arithmetic reuses that unit globe. It differs in three respects. Strength $r \leq 1$ is an activation of one arrow, not a purity parameter of a density matrix. Phase χ is an internal colour of that arrow, not a global $U(1)$ that can be quotiented away. There is

no measurement postulate: Mix may mint a third unit or black; $\uplus$ only collects. $\uplus$ never collapses a superposition onto an eigenstate. Related work may say: we reuse a unit globe; we do not borrow quantum authority for $\uplus$.

Word2vec and cosine embeddings. Lexical similarity as an angle is reused. A bag of words as a commutative Bundle is reused in spirit. What is added is a bounded unit ($r \leq 1$), an explicit collection operator that is idempotent ($A \uplus A = A$), and an order-dependent path T that cosine addition does not supply. Offset is a manifold cousin of the familiar king – man + woman arithmetic, not a proof that the toy globe is a full lexicon.

Transformers. A transformer occupies the same *slot* as T : it is a path through seated tokens. It is not the same device. The usual stack is

sentence $\rightarrow$ token $\rightarrow$ encode $\rightarrow$ embed $\rightarrow$ transformer.

The stack of this paper is

sentence $\rightarrow$ token $\rightarrow E_L \rightarrow$ sphere $\rightarrow \uplus$ (bowl) + T (walk).

Tokens and encodings stay. The embedding is a cast into one sphere (of whatever dimension the lexicon needs). A higher-order unit (a line, a report, a lexicon) may treat each information atom as the already-furnished stand-in for a token: that is a closure over many units, not a change to the atom. T_{poem} , T_{prose} , and T_{equation} are species of walk, not extra nets. A transformer may later fill those walks as a learned path. This paper does not claim to replace attention; it splits presence from order so that a learned path need not also invent the city.

RotatE. Knowledge-graph embeddings that treat a relation as a rotation in a complex plane are the nearest neighbour of T . Pattern Arithmetic extends that idea from S^1 (or a product of circles) to $\text{SO}(3)$ acting on S^2 , and pairs the rotation with an additive phase χ . It also keeps a separate Bundle operator, which RotatE does not.

von Mises–Fisher. Directional statistics on the sphere reuse the same stage. Those models estimate a density. Pattern Arithmetic does not: the atom is a point, r is not a concentration κ , and $\uplus$, $\oplus$, $\boxplus$, T , and offset are algebraic operations on units, not maximum-likelihood fits.

Riemann sphere. Stereographic compactification of $\mathbb{C} \cup \{\infty\}$ onto S^2 is the natural home of one numeric encoding (unbounded reals toward the poles). The algebra of this paper is not the Möbius group. A clock, or a stereo map, is a chart; + in **num** mode is decoded arithmetic, not a fractional linear map.

$\text{SO}(3)$. In three dimensions, the transformation T is a named action of the rotation group, plus a phase shift; beyond three, each T is a simple rotation — one plane turned, the rest held — and not a general element of $\text{SO}(N_{\text{dim}})$ (§4.4). That is the whole geometric engine of a path. What $\text{SO}(3)$ does not supply is the bowl, the cap $r \leq 1$, the sort τ , or Mix. Non-commutativity of poems is inherited; idempotence of $\uplus$ is not.

Spherical harmonics. A later field representation of many arrows could be stored as Y_ℓ^m coefficients, replacing an $O(N_{\text{vox}}^3)$ voxel cube by an $O(L^2)$ spectral array. That is a compression option for a field, not a definition of the atom, and it is not claimed as a result here.

11 Conclusion & Future Directions

The claim of this paper is narrow (Section 1.4). Counting is not the only arithmetic for combining units of information. A combination rule should match the unit. The sphere and the operators $\uplus$, $\oplus$, $\boxplus$, T , offset, and $\bowtie$ are one proposed encoding of that claim. They are not a general theory of language, physics, biology, or computation. The costumes, the enclosure, and related-work paragraphs are illustrations and boundaries, not domain models.

What was written is two pipelines on the same unit. Bundle many sorted units into a bowl of info atoms $x = (r, \hat{n}, \chi, \tau)$ on a unit sphere, then derive results by ordered transformation of a register occupant — or, where every contribution must survive intact rather than be walked or read, Collapse them into one counted atom instead. Separately, assemble units without fusing them and let external seats descend until a stated threshold freezes them; the frozen camps have bodies, not names. Naming a camp is left open here, on purpose: a second companion paper, *Pattern Arithmetic: Higher Truths from Assembly*, now states it as the question still to be answered, not a comparison already made. Embeddings, on this view, are not abolished. They are embeddings into one proposed common coordinate register, so that a concept such as man *may* keep the same angles under every lexicon that points at it. That is a design, not a demonstrated universal semantic space.

The spoken conventions, the mixed-document and shared-lexicon design, and the standardized dictionaries are carried by the companion paper on Inference, as seed and not as statute.

11.1 Future work

What remains to be built is a standardized concept dictionary together with per-language lexica E_L (companion paper, on Inference). Until those catalogues exist, tokenization and encoding remain the handles by which a paragraph finds its points. They may continue; they will only have to work lighter. The open bet is that they need not also remain the architects of a new space of meanings. The algebra closed in this paper does not produce a universal man point by itself. Standardization is sociological and empirical; it is not an axiom.

A further deferred question is whether a path T should stay hand-specified or carry learnable weights: the angle law $\alpha(r)$, a phase map, a gate between Mix ($\boxplus$) and T_{poem} , and the lexicon casts E_L . The honest constraint is structural. If undo-the-poem matters, learning should stay inside $\text{SO}(3)$, not Bundle T into an arbitrary matrix. Weights may change how the shells are thrown. They should not let the shells throw themselves.

Left fully open is the hypothesis raised in the table-and-chair costume of the companion paper: that separating presence ($\uplus$) from path (T) as distinct primitives should make higher-order truths cheaper to learn than recovering the same distinction from undifferentiated vectors. Testing this would need a controlled comparison — a compositional task, a flat-embedding baseline, and a T -path model, measured on data efficiency rather

than final accuracy — and no such comparison is attempted here. The table-and-chair and storm costumes motivate the hypothesis; they do not test it.

Ambient $\mathbb{R}^3$ makes the expressions $\mathcal{S} + (a, b, c)$ and $\mathcal{S} \cdot (a, b, c)$ look legal. They are legal as *vector* arithmetic. They are not automatically legal as *unit* arithmetic. A state with $r = 0.98$ added to another with $r = 0.98$ along the same direction is $r = 1.96$, which has left the ball. A scalar $2 \cdot (0.98)$ does the same. The unit cap $r \in [0, 1]$ then forces a choice, to be made later and not here: clip (overflow saturates at 1), project (renormalize direction and forget extra strength), or refuse (the operator is undefined off the ball). Schoolbook + on the `num` encoding is a different door: decode, add in $\mathbb{R}$, encode back if the chart allows. Do not let $\mathbb{R}^3$ addition quietly replace $\uplus$, $\oplus$, $\boxplus$, or T — $\oplus$ especially, since its own symbol is the one most easily mistaken for the plus sign it is not. The embedding lives in $\mathbb{R}^3$. The unit lives in the ball. The three-dimensional ball teaches the algebra; close document semantics, if pursued, live on a higher-dimensional sphere that carries the same verbs.

Cancellation planes. Take a plane through the origin that a path names. The vectors of an atom on either side of it give two sums, and the vectors lying in the plane are left out. The heights about the plane’s normal a balance when $a \cdot V_{\text{derived}} = 0$, which holds for every plane that contains the direction of V_{derived} ; every atom has such planes, and the statement says little. The stronger condition, that the vectors off the plane sum to zero, is a real one: an atom has such a plane exactly when its vectors split into a part that sums to zero and a part that lies in one plane through the origin. Which atoms have one such plane, several, or none, and whether those that have them share useful properties, is left open. It is recorded as a direction, not claimed, and no result of this paper depends on it.

Dimensionality. Nothing stated above is intrinsically three-dimensional; the stage may be $S^{N_{\text{dim}}-1}$ as well as S^2 . The magnitude cap $r \in [0, 1]$ is a property of a vector’s length, not of the ambient dimension. The log and exponential maps used for the offset operator (Equations around (30)–(31)) are already Riemannian statements that hold on any $S^{N_{\text{dim}}-1}$ in $\mathbb{R}^{N_{\text{dim}}}$, not only S^2 . $\text{SO}(3)$ generalizes to $\text{SO}(N_{\text{dim}})$ as an abstract group — associativity and invertibility are group-theoretic facts about $\text{SO}(N_{\text{dim}})$ for any N_{dim} , not special to $N_{\text{dim}} = 3$ — but that is a statement about the group, not about T_w ’s specific construction from one vector: §4.4 found that a single $\hat{n}_w$ does not carry enough data to fix a rotation outside three dimensions, however true the group’s own properties remain. Three dimensions were chosen for legibility and hand-checkable worked examples, not necessity. Raising N_{dim} is expected to relieve one specific problem: too few directions forcing distinct concepts to crowd onto nearby points, a kind of clipping by dimensional scarcity rather than by magnitude. Because crowding is a many-atom problem, the benefit is conditional on the bowl actually being large enough to crowd; raising N_{dim} for a bowl of two or three atoms relieves nothing, since ordinary distinct points already sit far apart on S^2 . It does *not* relieve the separate overflow problem discussed just above (clip, project, or refuse when r exceeds 1) — that is a magnitude question and stays open regardless of N_{dim} .

T and offset are no longer part of this TODO. §4.4 restates T for general N_{dim} , invertible and non-commutative up to $N_{\text{dim}} = 50$ and exact at $N_{\text{dim}} = 3$; §5.1 restates offset’s transport as a rotation within the plane of its two seats, exact at $N_{\text{dim}} = 3$. The remaining operators contain no step that depends on the dimension. Bundle is set

union. Mix reads phase among units that already share a direction. Assembly’s bracket $3 + \hat{n}_A \cdot \hat{n}_B$ stays within $[2, 4]$ in any dimension, so it never repels by the same argument as in three. The identities of [Axiom 7](#) are dot-product identities. Each follows from the argument just given, in any dimension; no run is needed to establish it. What stays three-dimensional is presentation: the polar chart (θ, ϕ) and every worked example, by the convention stated at the start of this paper. Mix across differing directions is refused on grounds stated once, for any dimension (§2.7); that refusal was always a question about direction, not dimension.

The algebra is dimension-blind and count-honest: every informational arrow is a unit; the stage may be S^2 or $S^{N_{\text{dim}}-1}$; packing more arrows into one 3-D atom is a packing choice, not a change of claim. 3-D with a high Q is the right teaching object. Higher N_{dim} is the same object with more sky — use it when crowding, not when the idea feels small.

Reducing a path by merging close actors: what the algebra guarantees, and where the rest belongs. Two *sequential* actors already compose into one equivalent rotation for free — that is just $\text{SO}(3)$ ’s (or $\text{SO}(N_{\text{dim}})$ ’s) closure, already latent in T ’s associativity. Replacing several merely *close* actors — by some stated distance on $\mathcal{S}$, not identical — with one representative before a path is walked is a different operation entirely: a coarse-graining step, not composition, and this paper states only its shape, not its technique. Any such reduction produces an additional, derived actor for a shorter path to use — it does not remove the original close actors from the bowl. The bowl under $\uplus$ (Section 3.1) already keeps every distinct seated unit; a reduced actor sits alongside its sources, not in place of them, and the unreduced path through the original actors remains walkable and exact. Reduction is a second, optional reading of the same bowl, the way Mix and T already are, not a rewrite of what the bowl holds.

Which specific merge rule to use — an average, a set of representative regimes, or a reduction that preserves the walk’s own shape at reduced length — and what distance and drift bound should govern it, depend entirely on what the reduction is for. That choice of purpose belongs to Inference’s own reference-atom and classification machinery, not to this paper’s algebra, and is not prescribed here.

Pair	+	$\uplus$	$\oplus$	$\boxplus$	$T_w(x)$	$x - a + b$
(word , word)	no	yes	yes	yes [†]	yes	yes
(num , num)	yes	yes [‡]	yes [¶]	no	no*	no*
(word , num)	no	yes [‡]	yes [¶]	no	typed	no
(num , word)	no	yes [‡]	yes [¶]	no	typed	no
(color , color)	no	yes	yes	yes	yes	open
(note-pitch , note-pitch)	no	yes	yes	yes	yes	open
(chladni , chladni)	no	yes	yes	yes [§]	open	open
(τ_1, τ_2) any other	no	yes [‡]	yes [¶]	open	open	open

Table 2: Sort discipline on a single sphere. Every $\boxplus$ entry is additionally subject to the place restriction of [Axiom 5](#): Mix combines χ at a common $\hat{n}$, and Mix across differing directions is refused (Section 2.7). **color** and **note-pitch** are the Tier-1 sorts of Table 1; they Bundle like any other sort, and their Mix is a domain reading rather than inert geometry. Mix on (**num**, **num**) is the clock trap of Section 6.2: it is geometrically defined and type-refused. [†]Mix on (**word**, **word**) is allowed as geometry and carries no linguistic χ -claim ([Axiom 0](#), Tier 0). *Numeric T and numeric offset may be defined later as maps of the chosen numeric encoding; they are not the lexical operators of Sections 4 and 5. Offset on the Tier-1 sorts is untested here and marked open rather than claimed. [‡]Bundle is sort-blind: the $\uplus$ column is *yes* for every pair, including pairs of different sorts and sorts not listed here. A bowl records which units are present, and presence carries no domain claim, so there is nothing for a sort mismatch to corrupt. What Bundle does *not* do is combine: $2 \uplus \mathbf{two} = \{2, \mathbf{two}\}$ is two seats and no cast, and $1 \uplus 1 = \{1\}$ is one seat and not the sum 2. Counting stays with $+$ (Section 6.2). The final row is the general rule; a sort seated on the sphere may always be Bundled, while its other operators wait on [Axiom 0](#). [¶]Collapse, like Bundle, is *yes* for every pair, of one sort or of two: it keeps every core exactly as it was and computes nothing, so a sort mismatch has nothing to corrupt, and (**num**, **num**) is *yes* because Collapse never computes the clock-trap sum that refuses Mix there. It differs from Bundle in keeping the count. A Collapsed atom of several sorts claims no relation among them ([Axiom 0](#)); each vector carries its own τ_i (Axiom 7). A sort inadmissible for Tier 1, such as **snowflake** in Table 1, still falls under this general row at Tier 0: it may be Bundled or Collapsed, and its other operators remain open until a stated map is given — the same status Table 1 already grants it, not a stricter one. What has genuinely not been done for **snowflake**, unlike **num**, **word**, **color**, or **chladni**, is the construction of an explicit chart; Tier-0 admission is asserted here, not built. A sort with *no* row in either table — one this paper has not considered at all — is undecided in the stronger sense of Table 1’s closing line, and is not seated until one is given. [§]Mix on (**chladni**, **chladni**) is a domain reading only within one mode, where the place restriction above is satisfied automatically; across modes it is undefined rather than merely untested (Section 2.7). T is marked open because direction carries a mode label for this sort and no domain meaning is claimed for rotating it.